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Lee et al.

(10) **Patent No.: US 12,400,911 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THEREOF**

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(73) Assignee: **Samsung Electronics Co., Ltd.** (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 291 days.

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(22) Filed: **Sep. 19, 2022**

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(30) **Foreign Application Priority Data**

Dec. 1, 2021 (KR) 10-2021-0169761

(51) **Int. Cl.**

H01L 21/768 (2006.01)
H01L 23/522 (2006.01)
H01L 23/532 (2006.01)
H10D 64/66 (2025.01)

(52) **U.S. Cl.**

CPC .. ***H01L 21/76811*** (2013.01); ***H01L 21/76877*** (2013.01); ***H01L 23/5226*** (2013.01); ***H01L 23/53266*** (2013.01); ***H10D 64/671*** (2025.01)

(58) **Field of Classification Search**

CPC combination set(s) only.
See application file for complete search history.

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Primary Examiner — Thanhha S Pham
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(57) **ABSTRACT**

The present disclosure provides a semiconductor device capable of improving element performance and reliability. The semiconductor device comprises a lower wiring structure, an upper interlayer insulating layer disposed on the lower wiring structure and including an upper wiring trench, the upper wiring trench exposing a portion of the lower wiring structure, and an upper wiring structure including an upper liner and an upper filling layer on the upper liner in the upper wiring trench, wherein the upper liner includes a sidewall portion extending along a sidewall of the upper wiring trench and a bottom portion extending along a bottom surface of the upper wiring trench, the sidewall portion of the upper liner includes cobalt (Co) and ruthenium (Ru), and the bottom portion of the upper liner is formed of cobalt (Co).

20 Claims, 29 Drawing Sheets

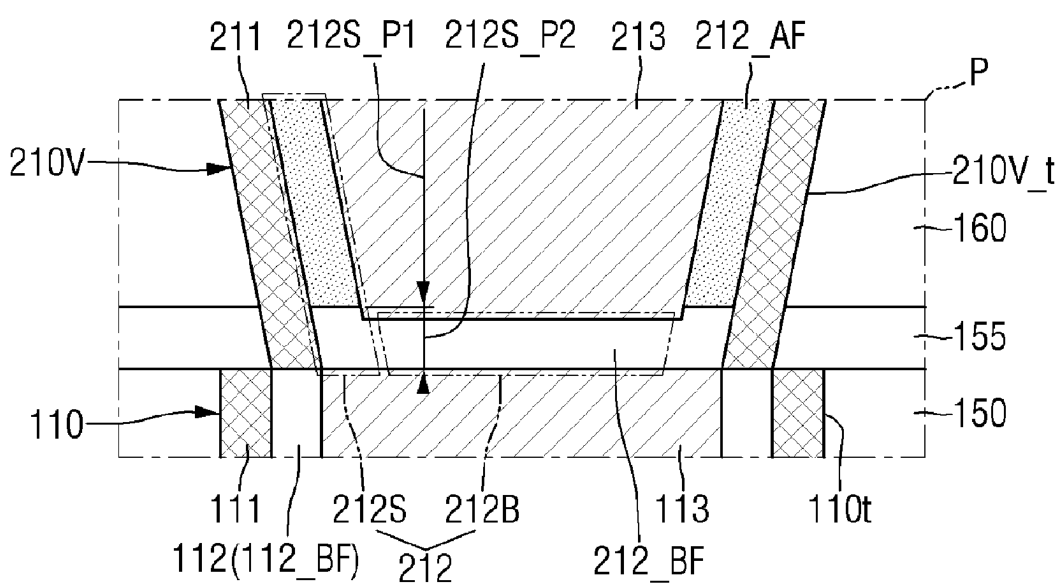

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FIG. 1

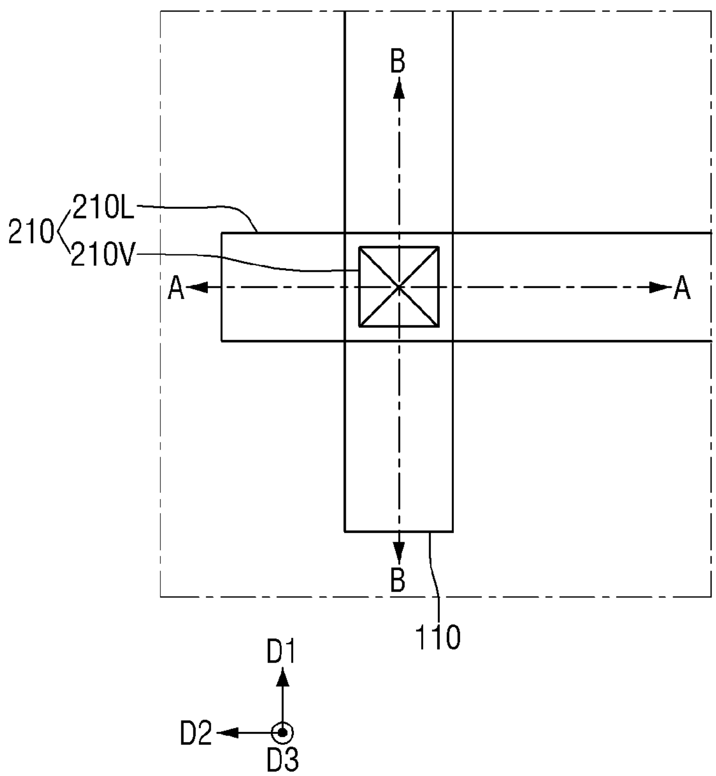
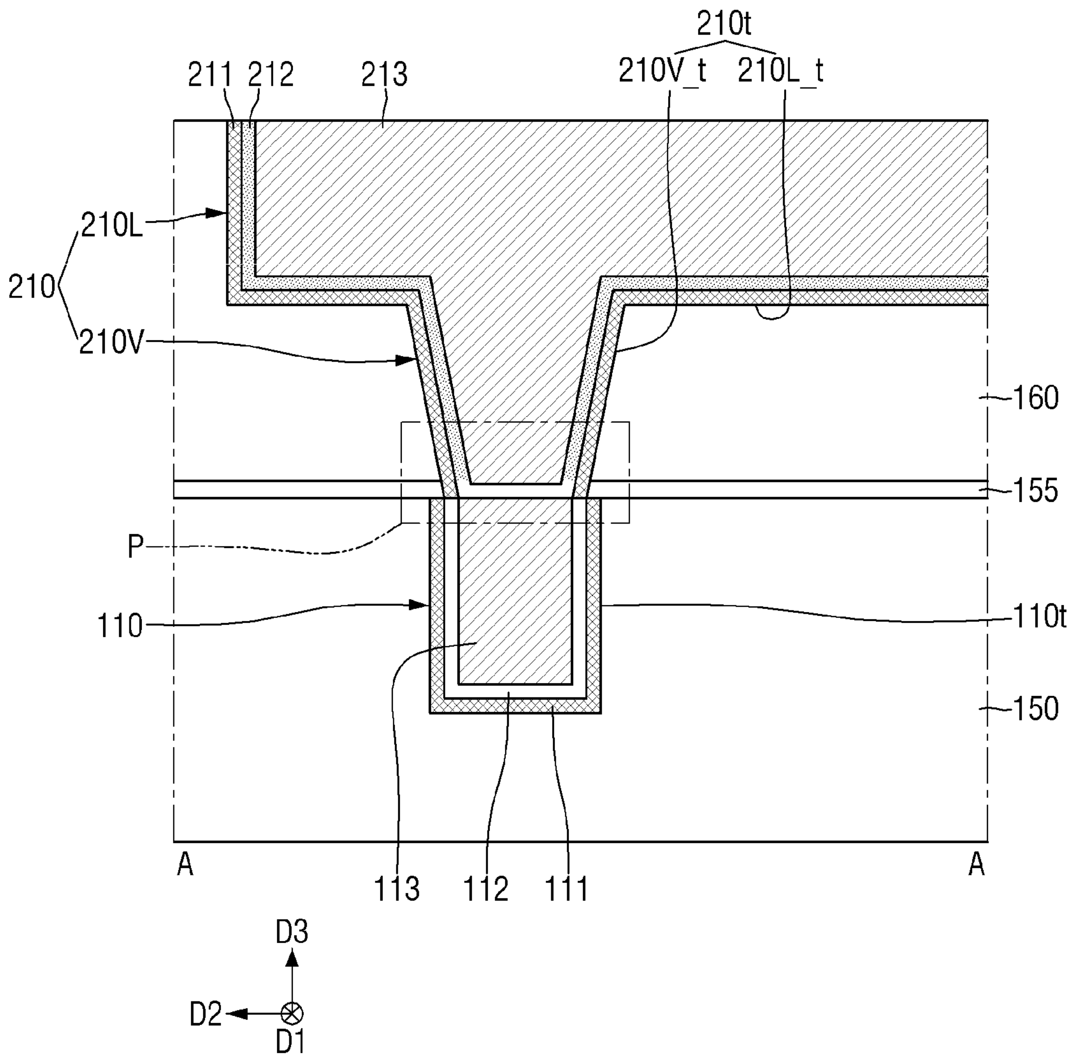


FIG. 2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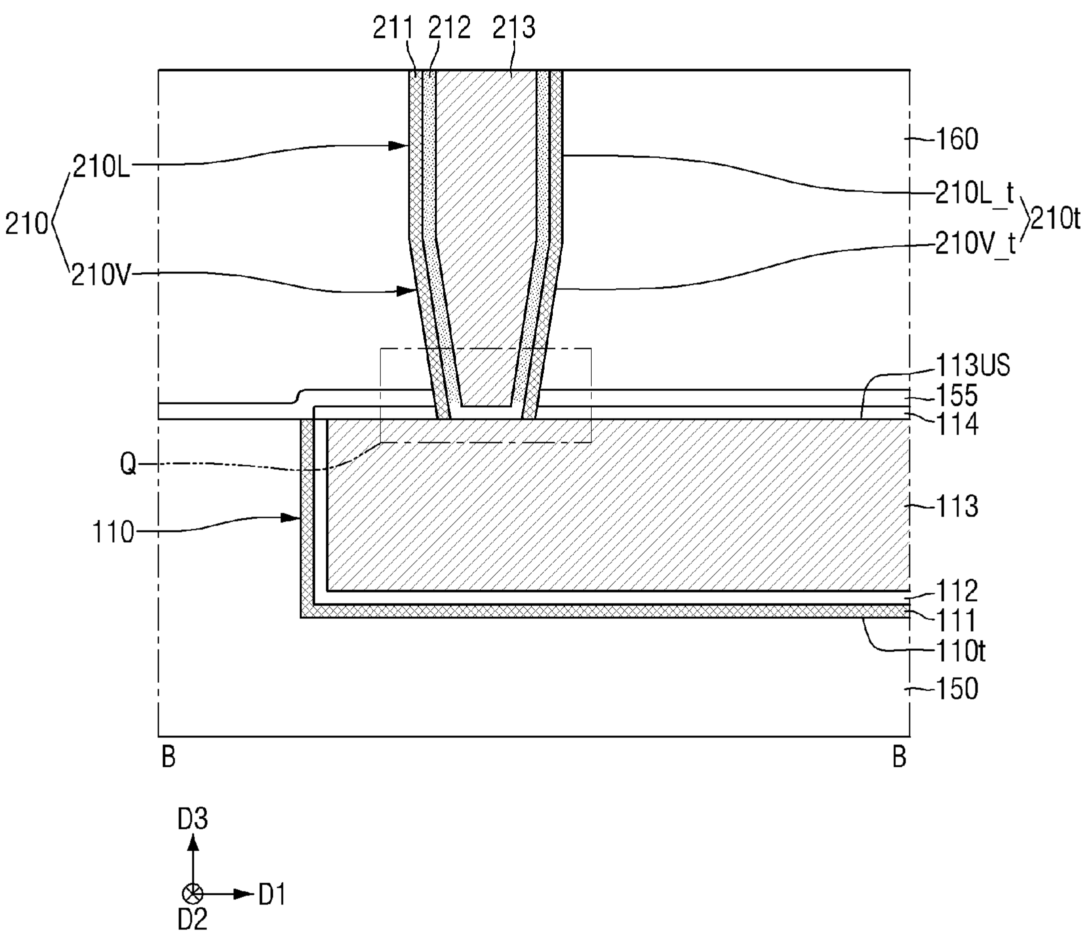
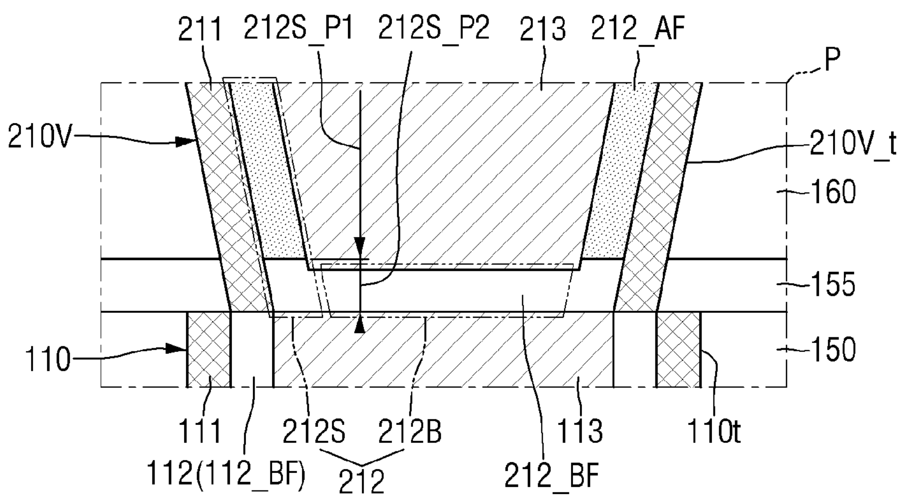
210L_t
210L
210
210V
160
155
P
110
110t
150
A
A
113
112
111
D3
D2
D1

FIG. 3



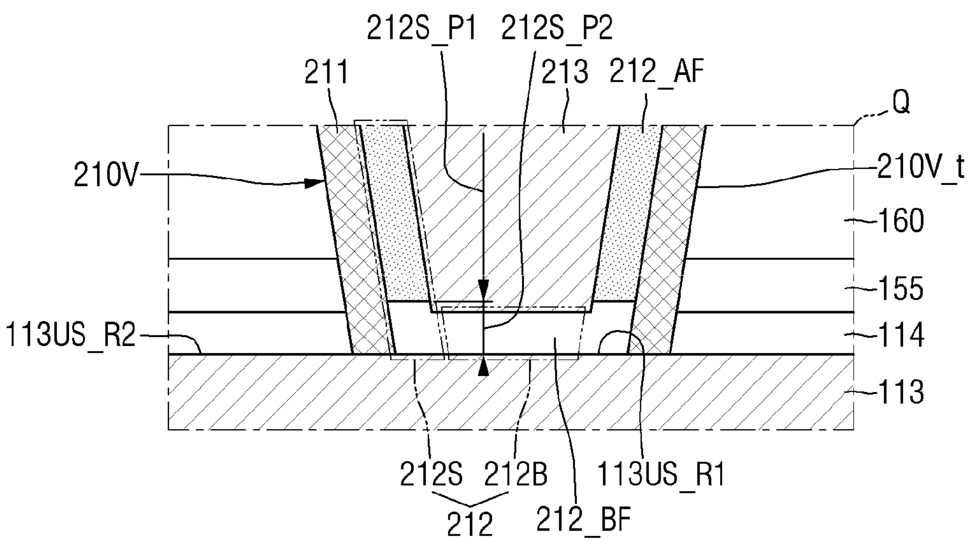
211
212
213
160
210L
210L_t
210t
210
210V
210V_t
113US
155
114
Q
113
110
112
111
110t
150
B
B
D3
D1
D2

FIG. 4



211
212S_P1
212S_P2
213
212_AF
P
210V
210V_t
160
155
110
150
111
212S
212B
113
110t
112(112_BF)
212
212_BF

FIG. 5


212S_P1
212S_P2
211
213
212_AF
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210V
210V_t
160
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113US_R2
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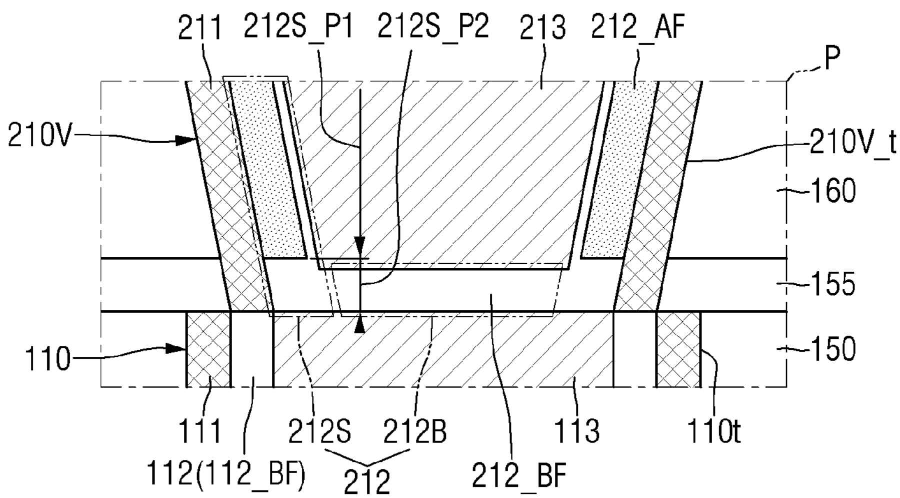
113
212S
212B
113US_R1
212
212_BF

FIG. 6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212S_P1
212S_P2
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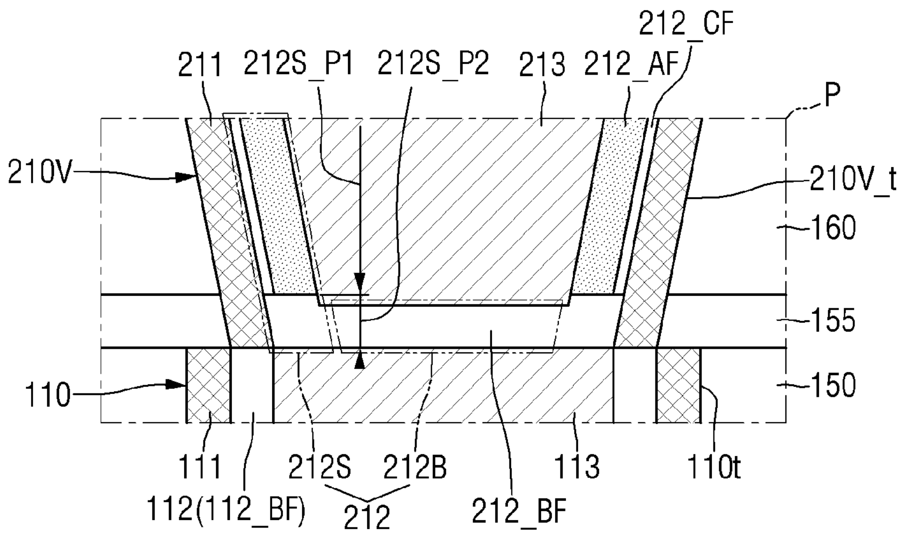
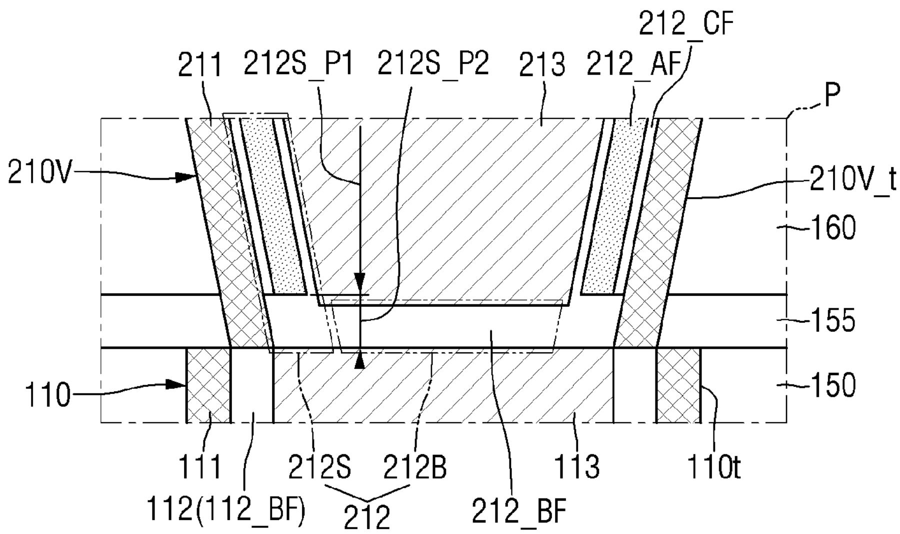
212_AF
P
210V
210V_t
160
155
110
150
111
112(112_BF)
212S
212B
212
113
212_BF
110t

FIG. 7



212_CF
211
212S_P1
212S_P2
213
212_AF
P
210V
210V_t
160
155
110
150
111
112(112_BF)
212S
212B
212
113
212_BF
110t

FIG. 8



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212S_P1
212S_P2
213
212_AF
P
210V
210V_t
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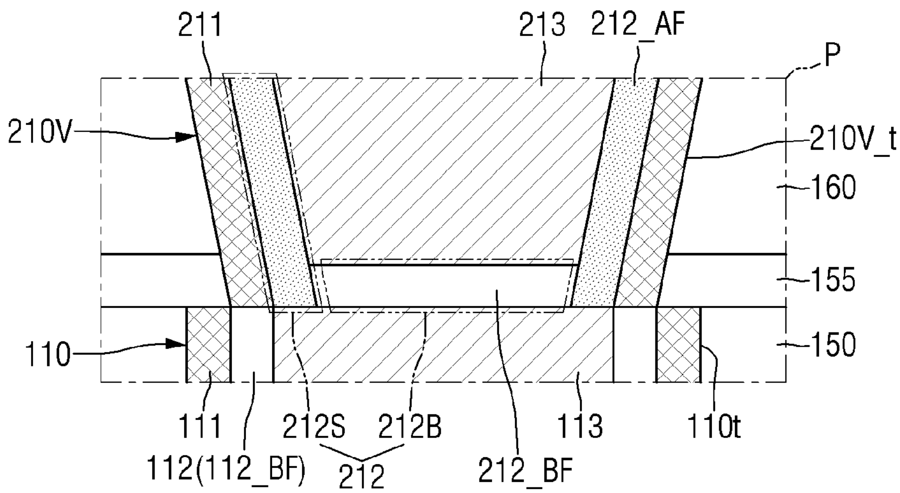
155
110
150
111
112(112_BF)
212S
212B
212
113
212_BF
110t

FIG. 9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213
212_AF
P
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210V_t
160
155
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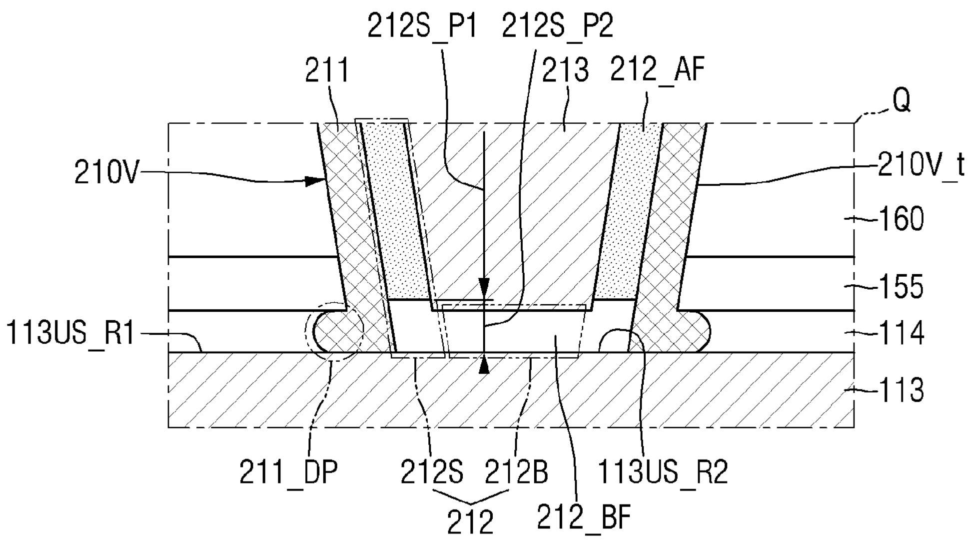
112(112_BF)
212S
212B
212
113
110t
212_BF

FIG. 10


212S_P1
212S_P2
211
213
212_AF
Q
210V
210V_t
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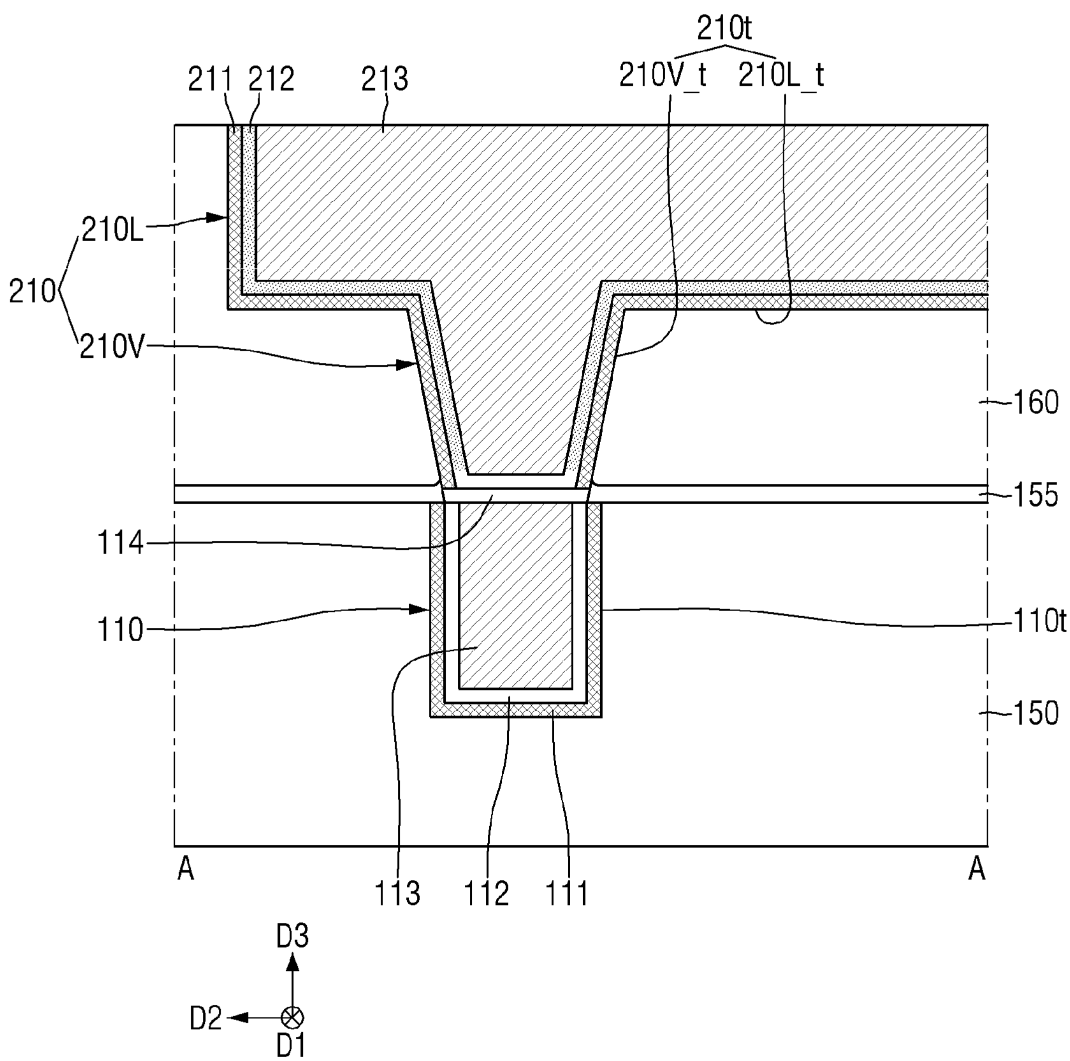
113US_R1
114
113
211_DP
212S
212B
113US_R2
212
212_BF

FIG. 11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210V_t
210L_t
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210V
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155
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110
110t
150
A
A
113
112
111
D3
D2
D1

FIG. 12

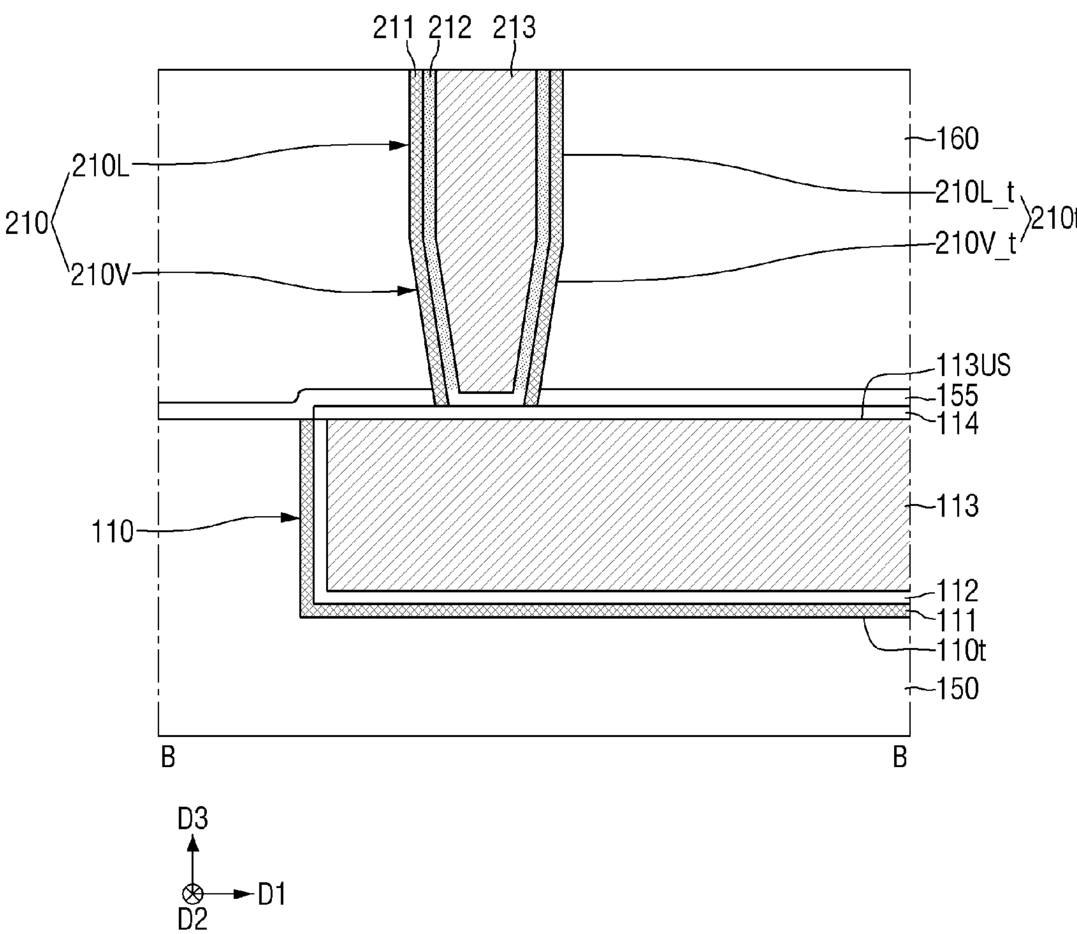
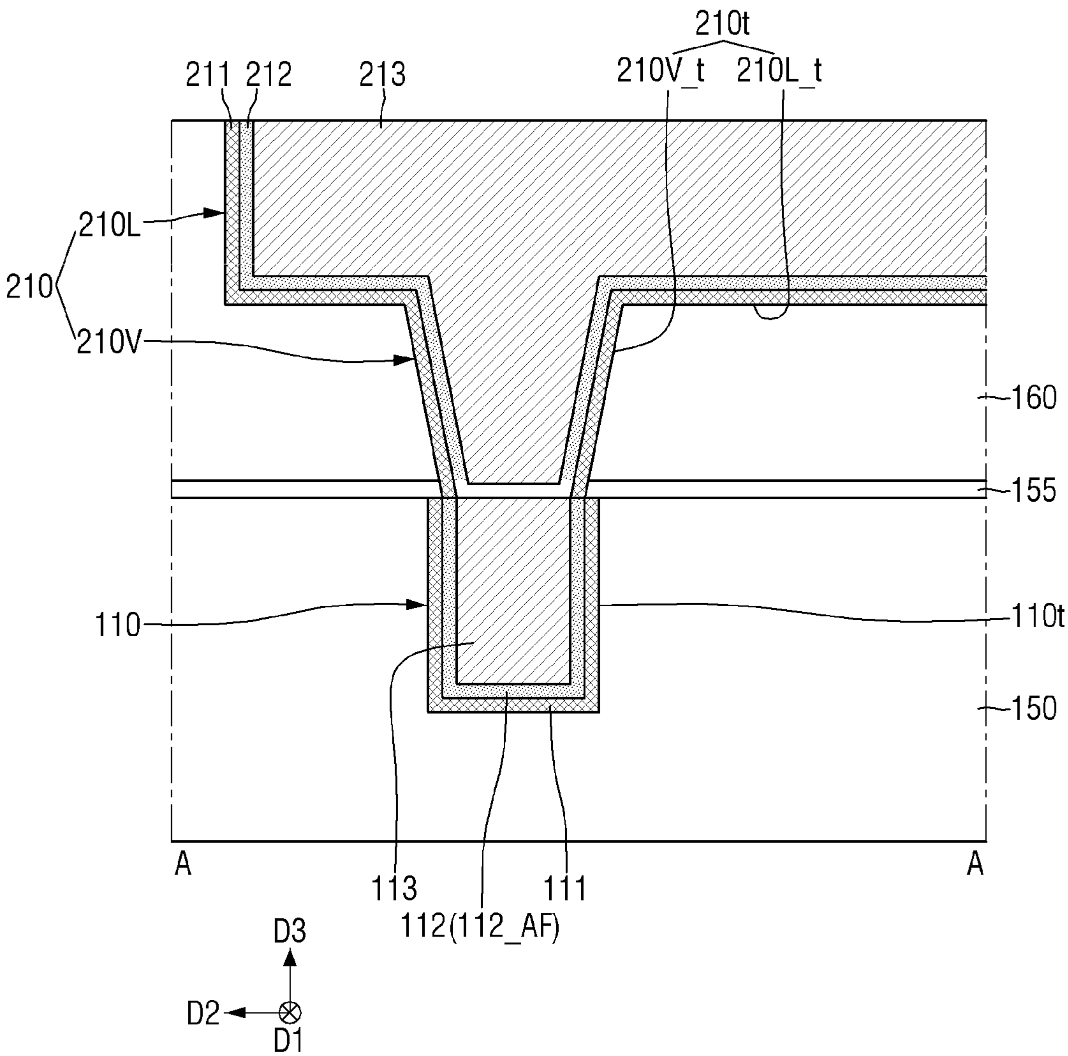

211 212 213
160
210L
210
210V
210L_t
210V_t
210t
113US
155
114
113
110
112
111
110t
150
B
B
D3
D1
D2

FIG. 13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211
212
213
210V_t
210L_t
210L
210
210V
160
155
110
110t
150
A
A
113
111
112(112_AF)
D3
D2
D1

FIG. 14

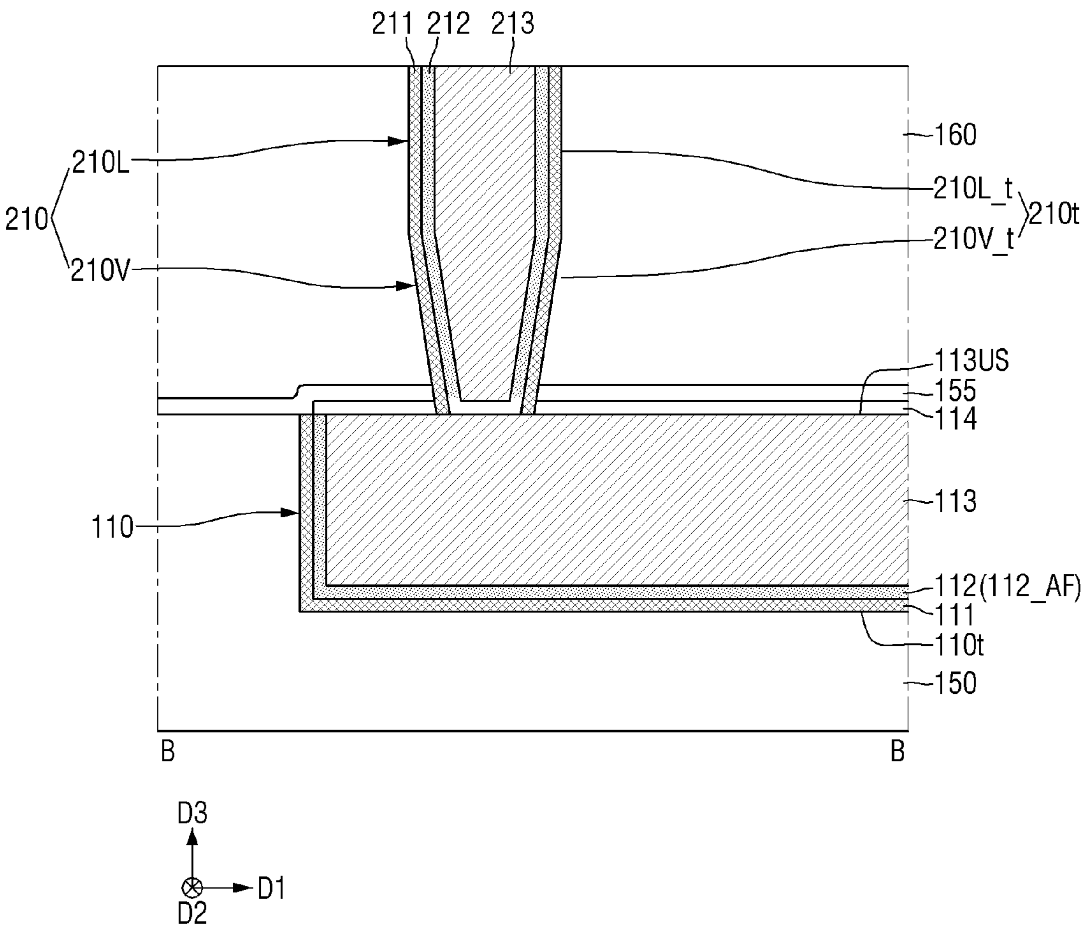
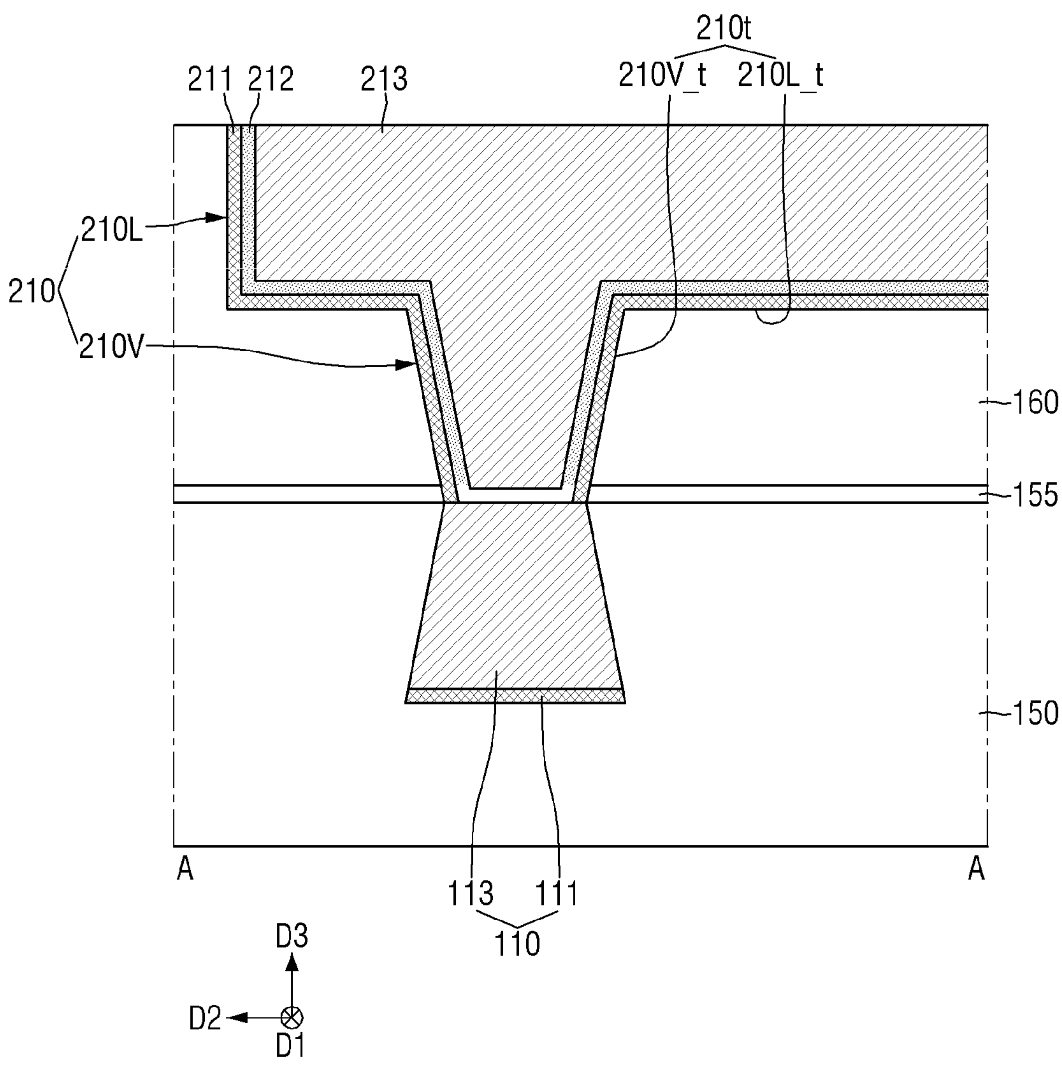
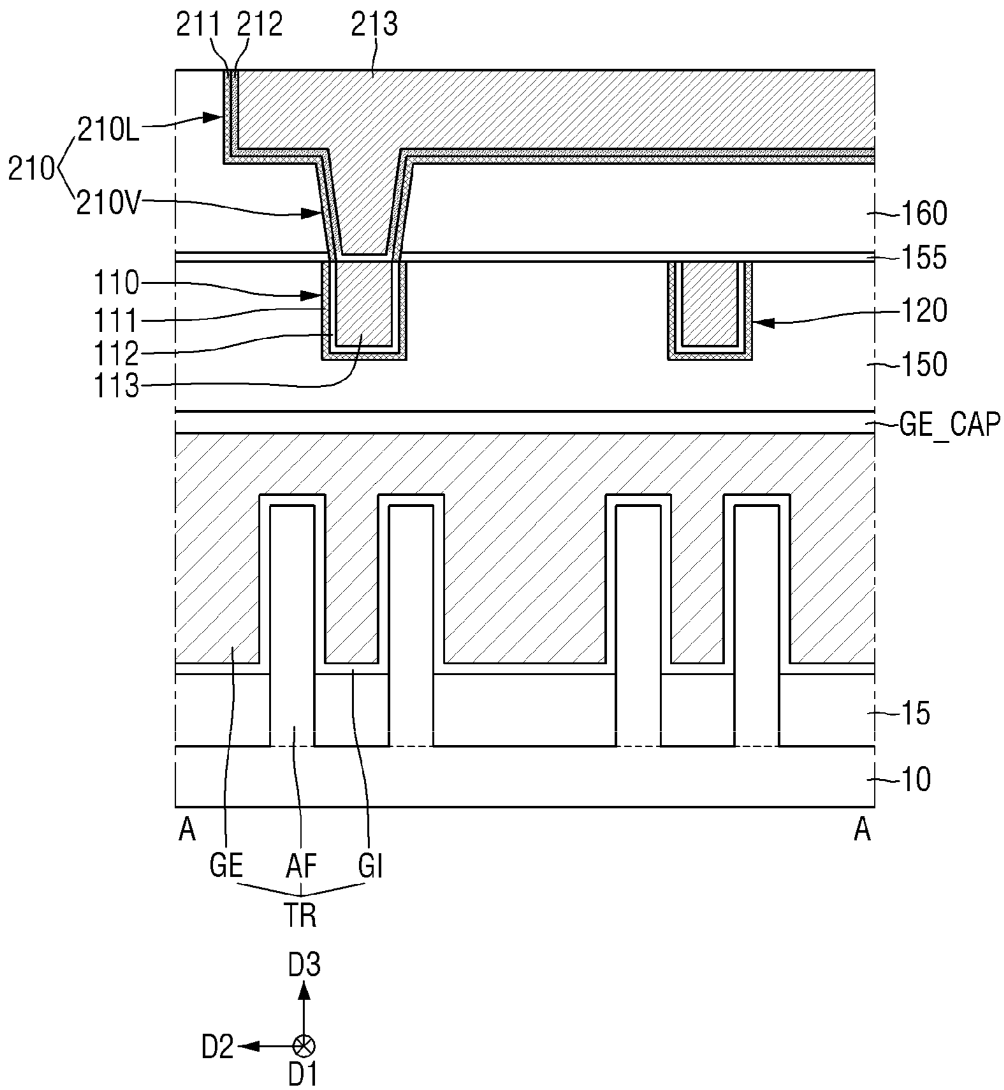

211
212
213
160
210L
210L_t
210
210t
210V
210V_t
113US
155
114
113
110
112(112_AF)
111
110t
150
B
B
D3
D1
D2

FIG. 15



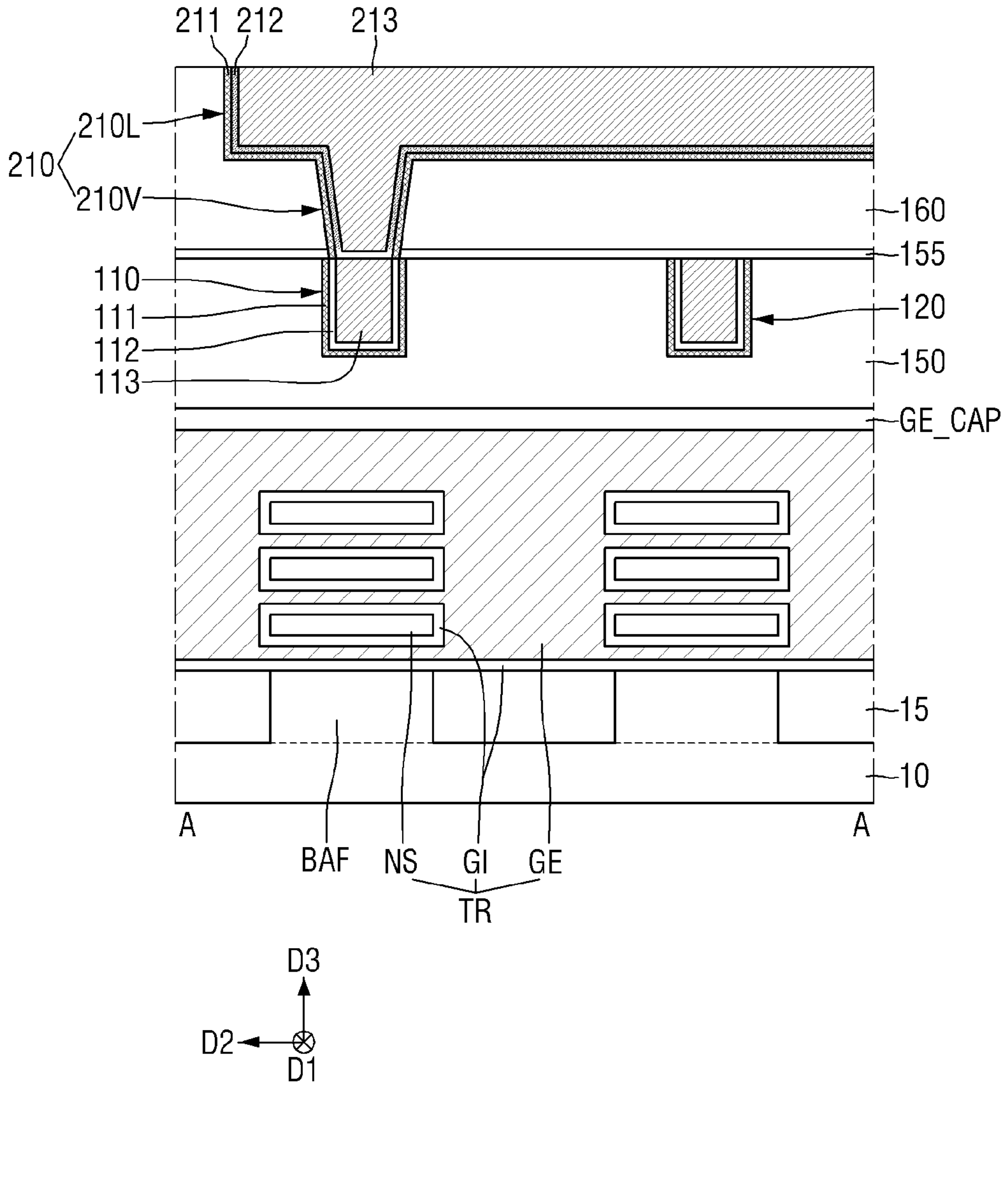
210t
211
212
213
210V_t
210L_t
210L
210
210V
160
155
150
A
A
113
111
110
D3
D2
D1

FIG. 16



211
212
213
210
210L
210V
160
155
110
111
112
113
120
150
GE_CAP
15
10
A
A
GE
AF
GI
TR
D3
D2
D1

FIG. 17



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212
213
210L
210
210V
160
155
110
111
112
113
120
150
GE_CAP
15
10
A
A
BAF
NS
GI
GE
TR
D3
D2
D1

FIG. 18

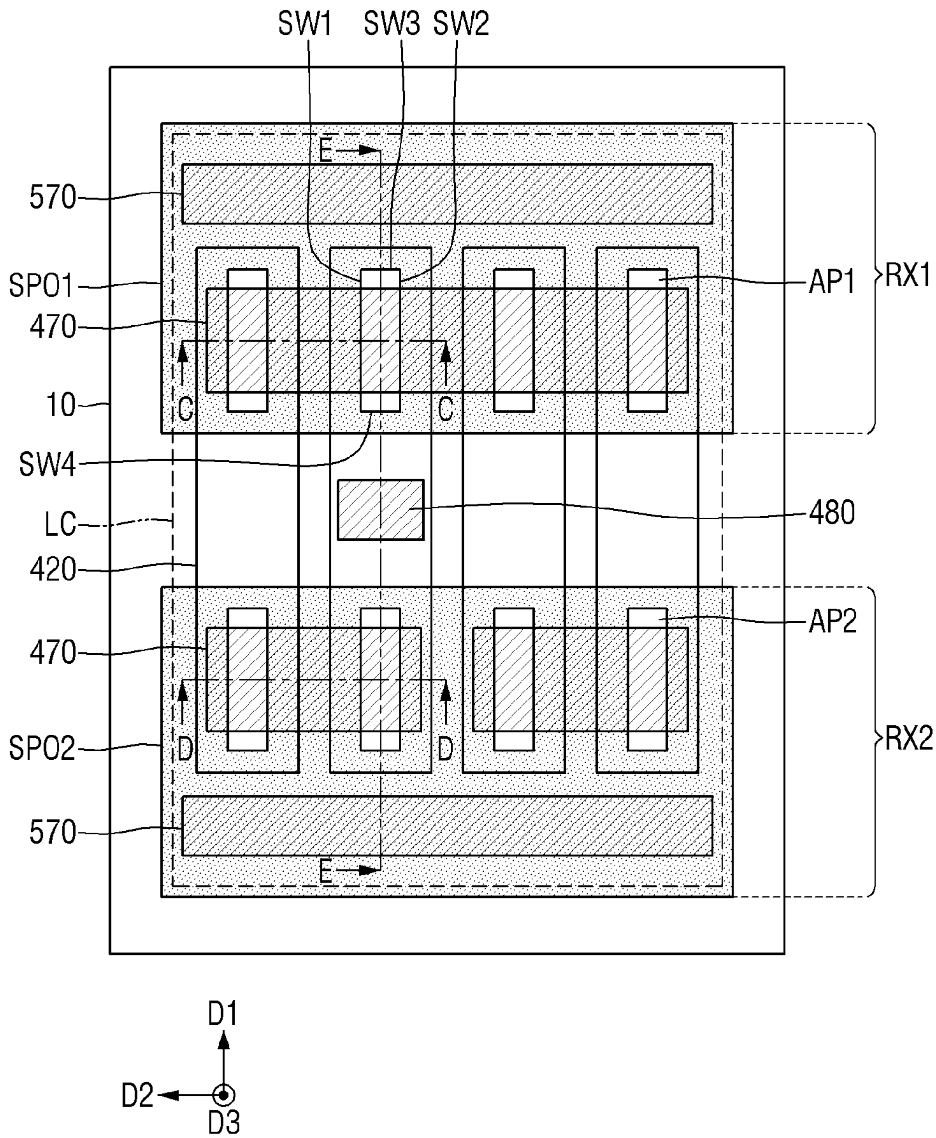

SW1
SW3
SW2
E
570
SPO1
470
10
C
C
SW4
LC
420
480
AP1
RX1
AP2
470
SPO2
D
D
RX2
570
E
D1
D2
D3

FIG. 19

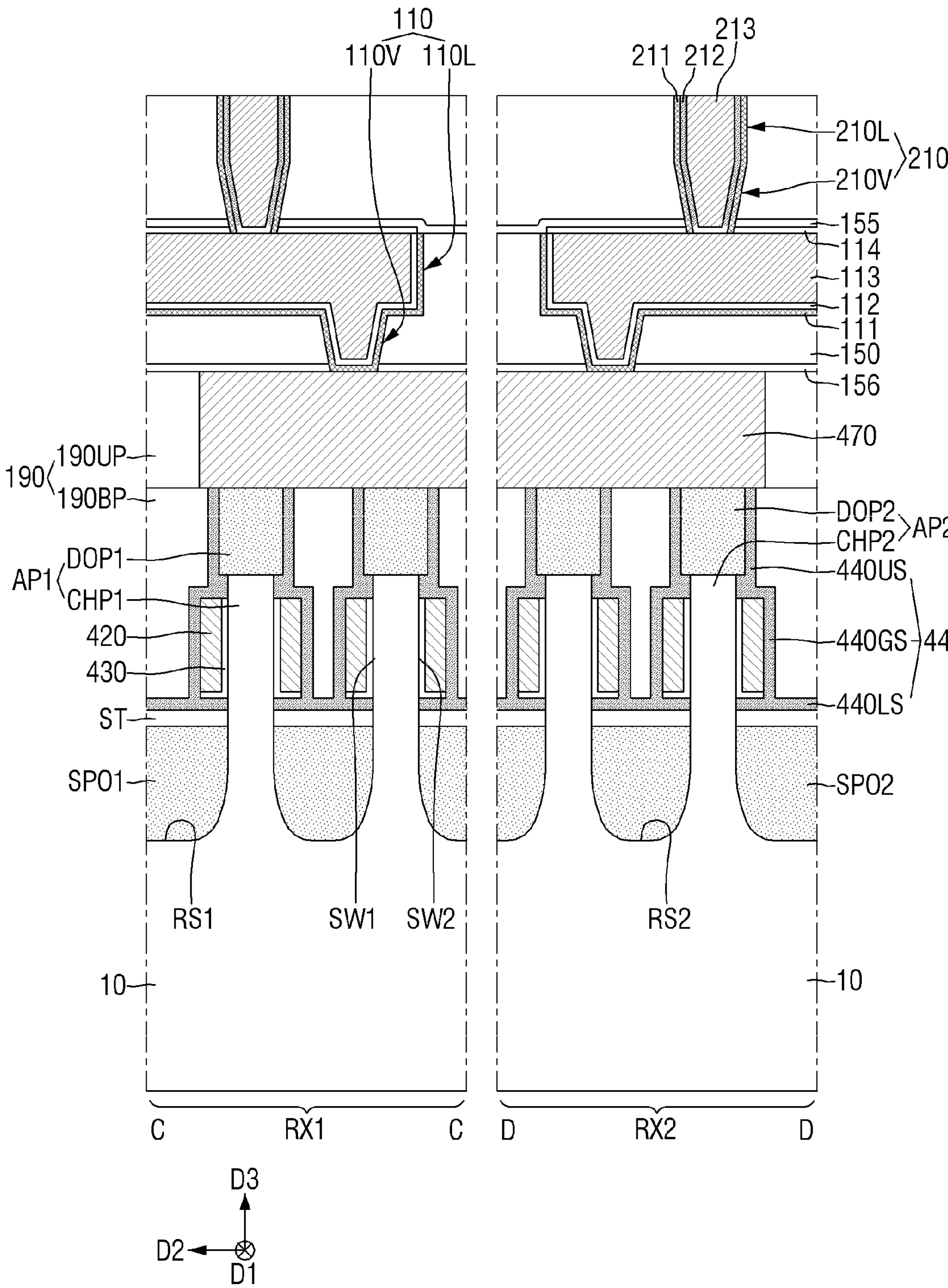
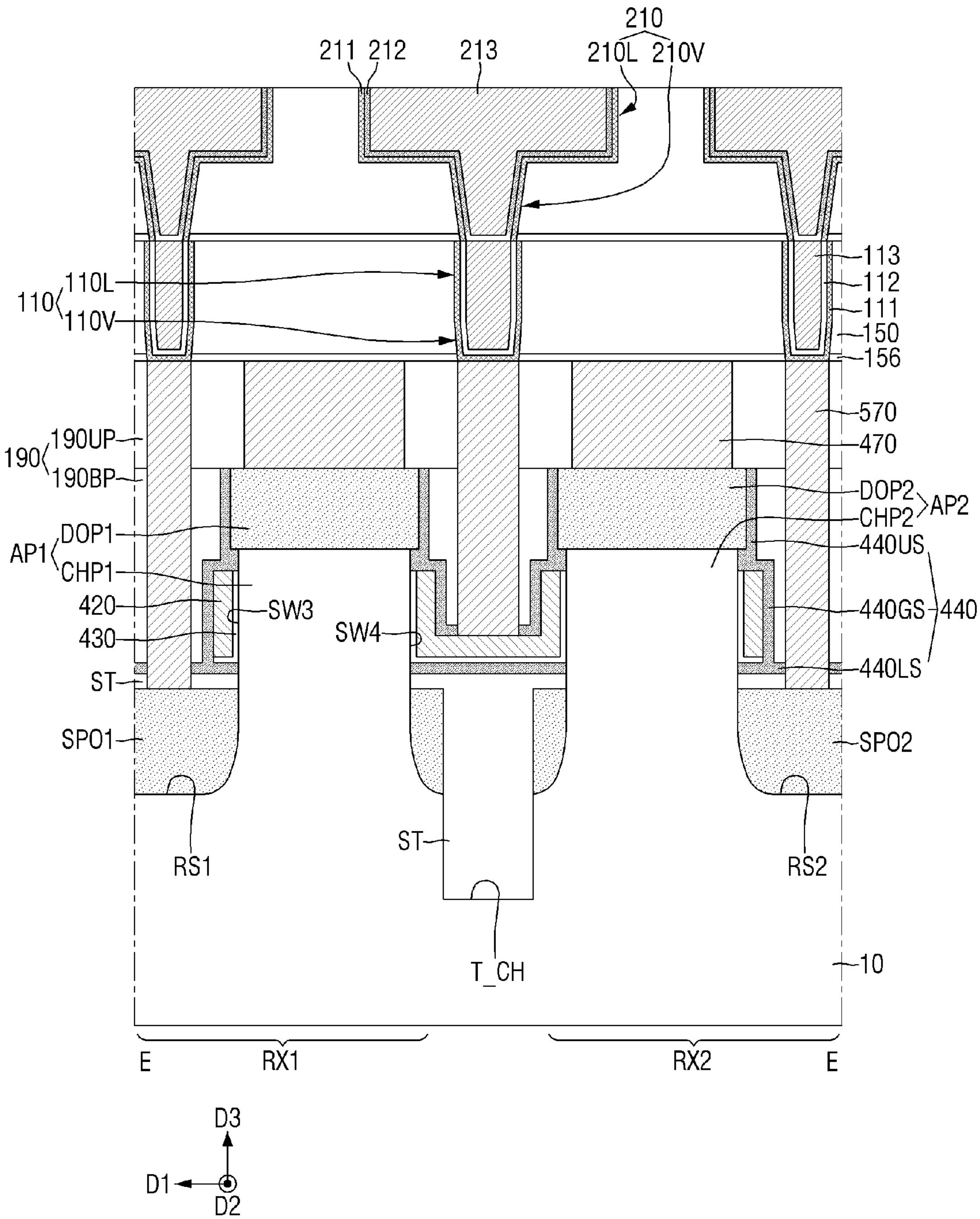
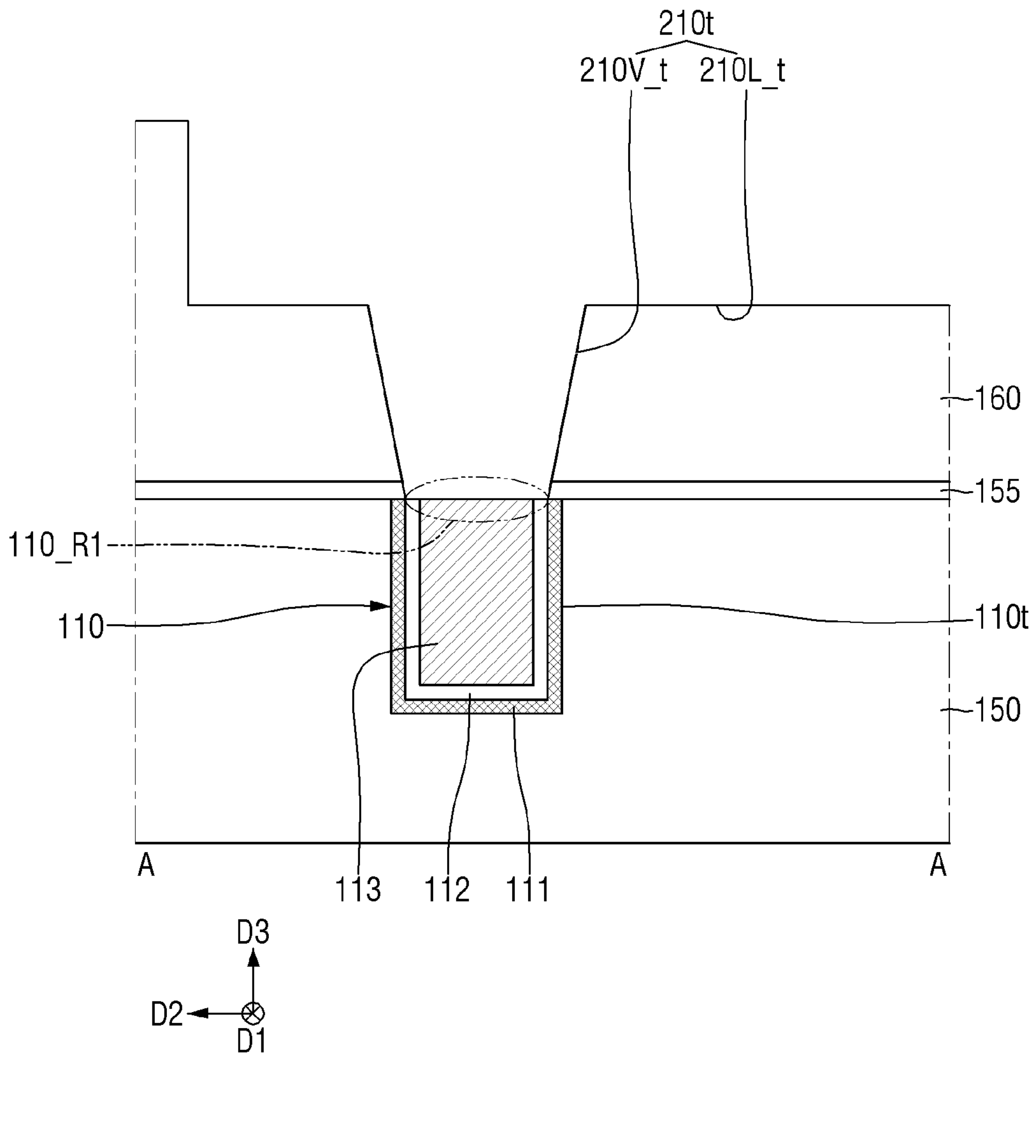

110
110V
110L
211
212
213
210L
210
210V
155
114
113
112
111
150
156
470
190UP
190
190BP
DOP2
AP2
CHP2
440US
DOP1
AP1
CHP1
420
440GS
440
430
ST
440LS
SPO1
SPO2
RS1
SW1
SW2
RS2
10
10
C
RX1
C
D
RX2
D
D3
D2
D1

FIG. 20



210
211
212
213
210L
210V
113
112
111
150
156
110
110L
110V
570
470
190
190UP
190BP
DOP2
CHP2
AP2
AP1
DOP1
CHP1
440US
420
SW3
430
SW4
440GS
440
440LS
ST
SPO1
SPO2
RS1
ST
RS2
T_CH
10
E
RX1
RX2
E
D3
D1
D2

FIG. 21



210t
210V_t
210L_t
160
155
110_R1
110
110t
150
A
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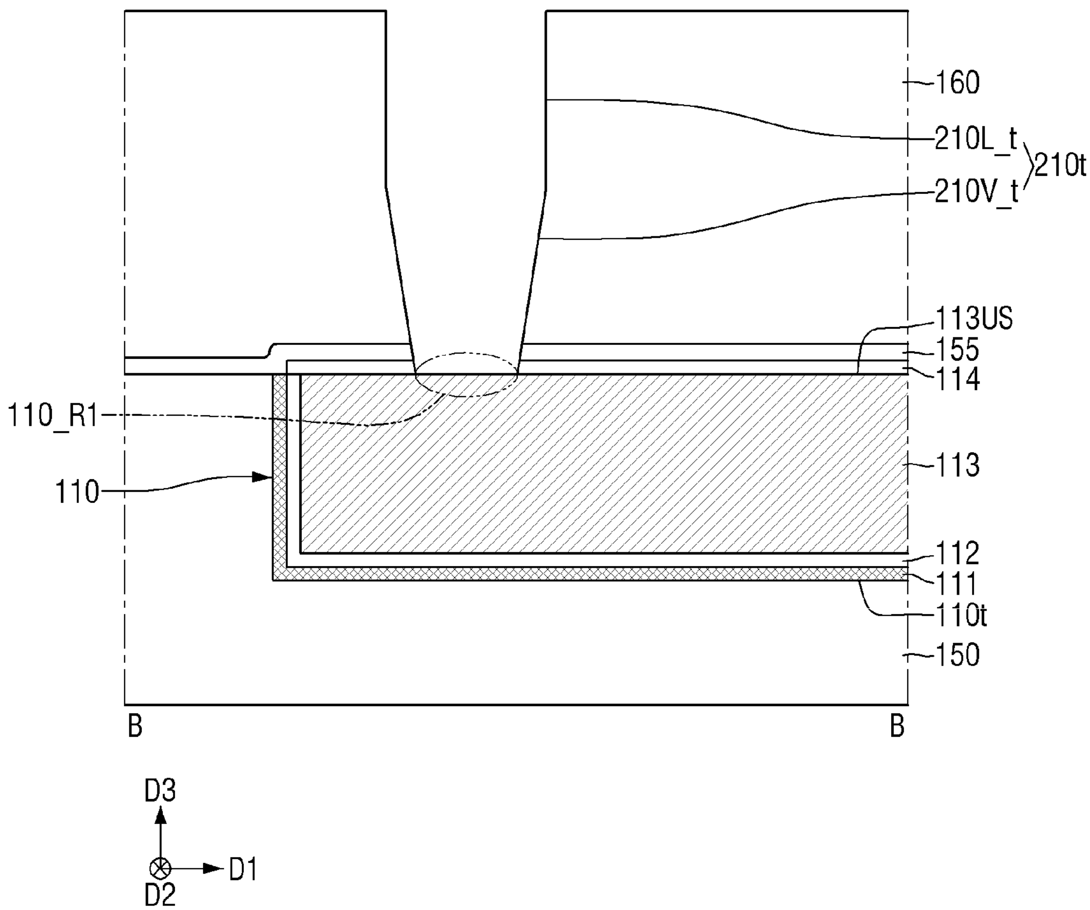
113
112
111
D3
D2
D1

FIG. 22


160
210L_t
210t
210V_t
113US
155
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110_R1
110
113
112
111
110t
150
B
B
D3
D1
D2

FIG. 23

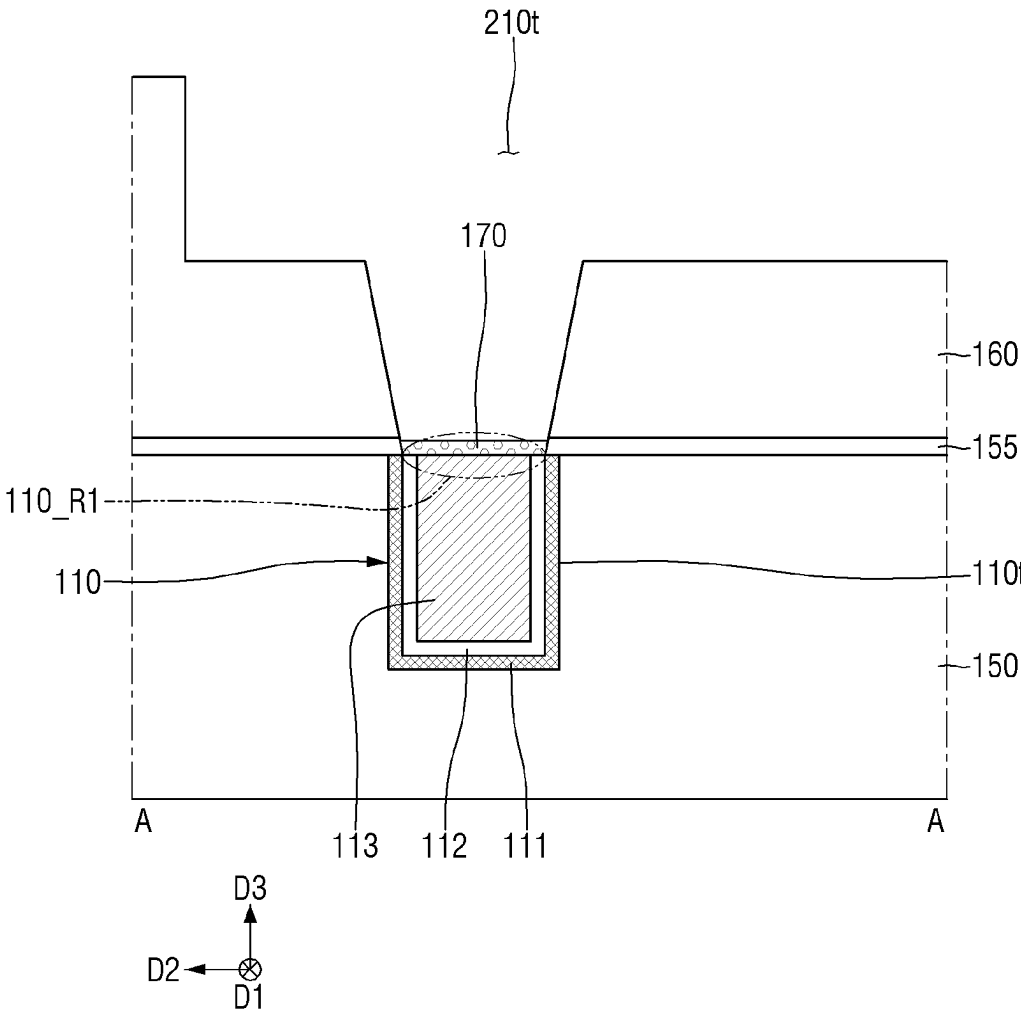
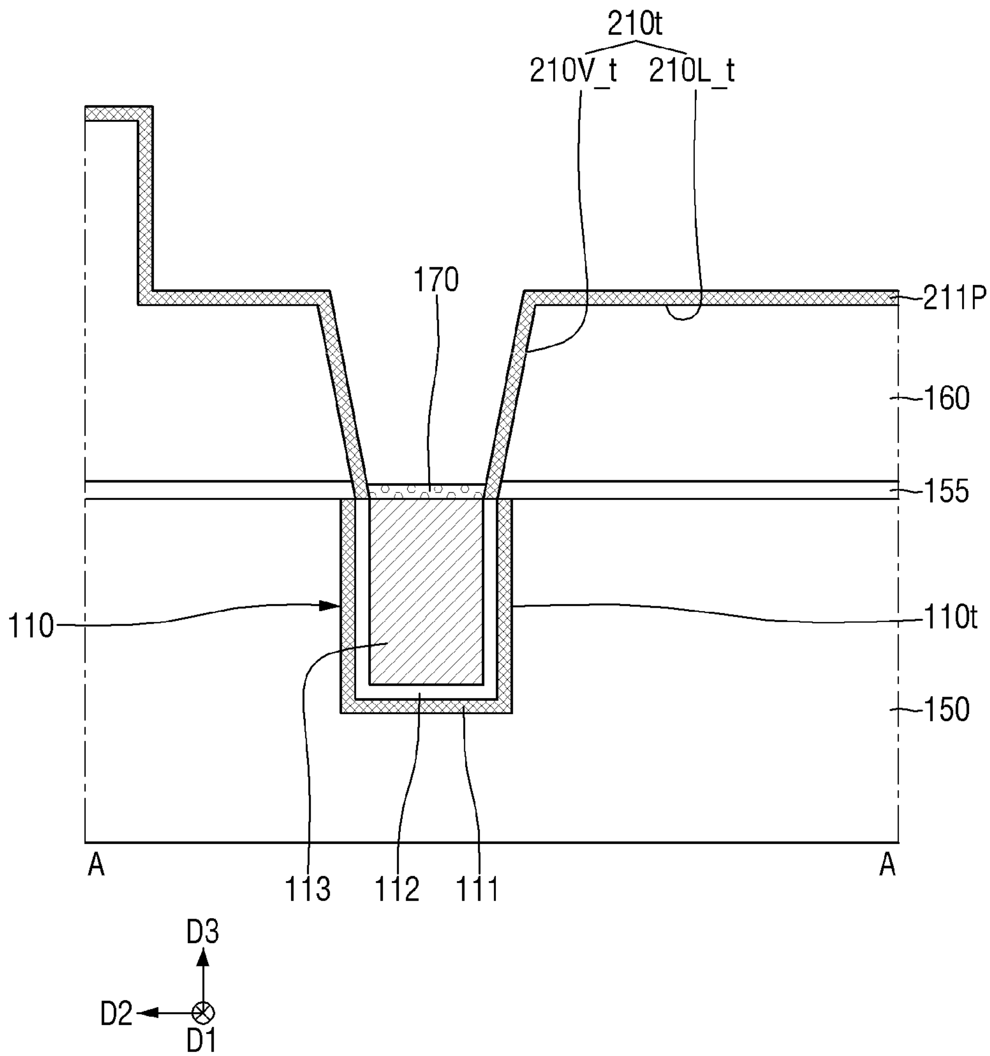


FIG. 24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210V_t
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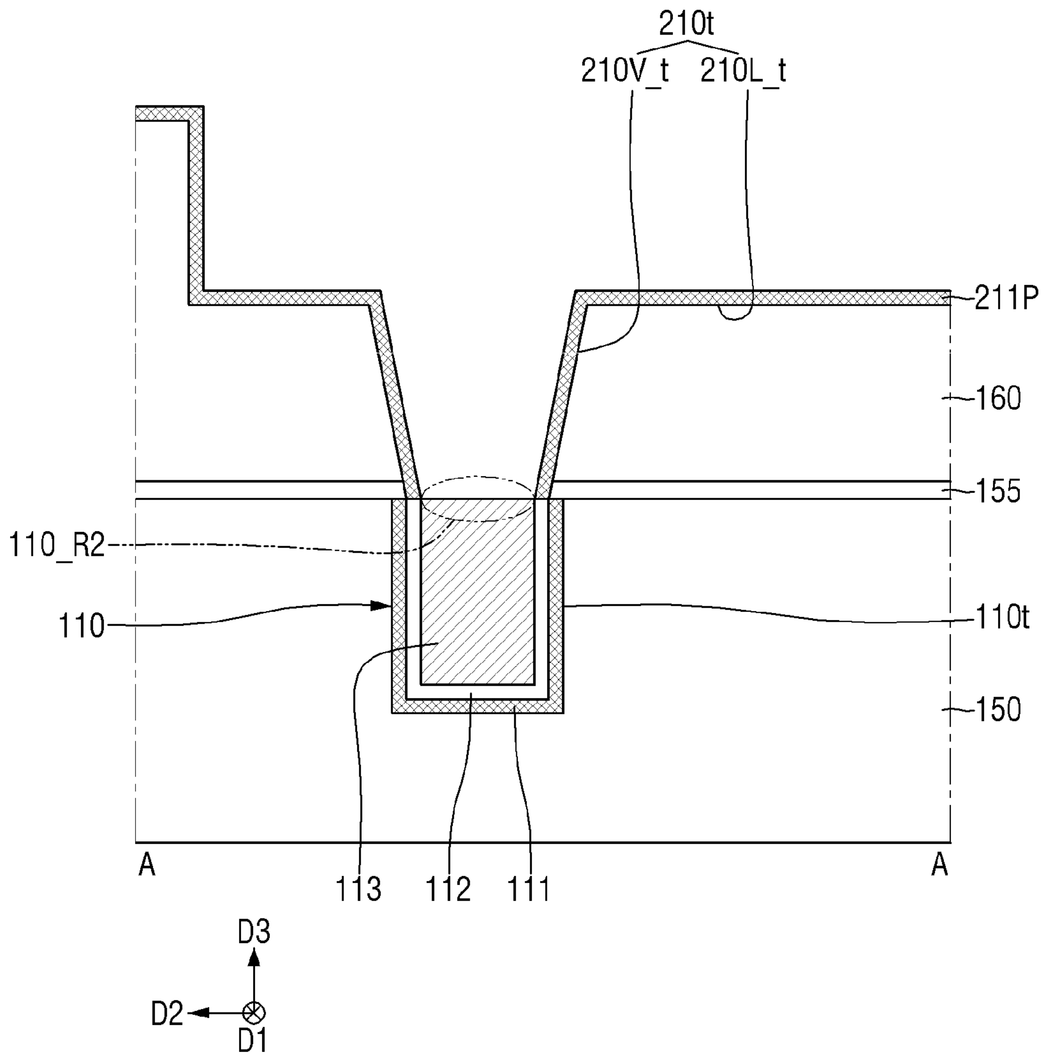
D3
D2
D1

FIG. 25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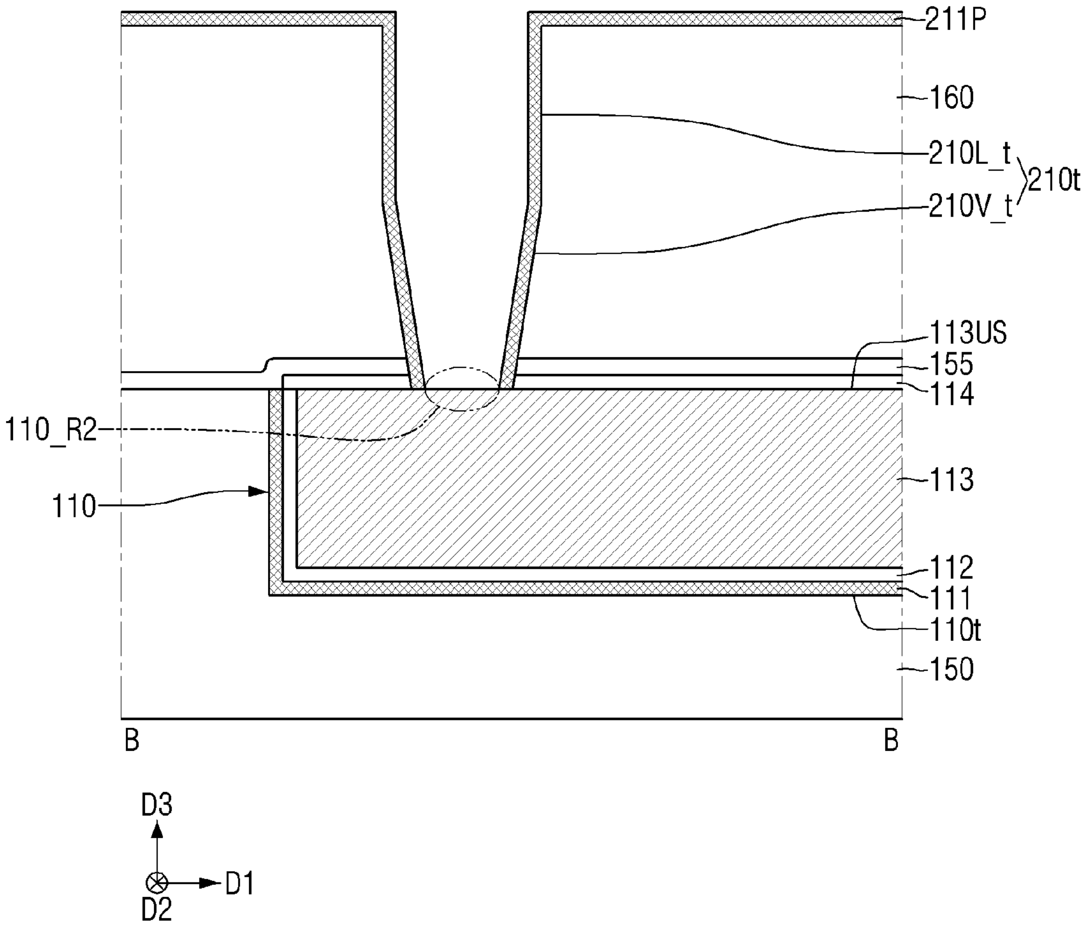
D3
D2
D1

FIG. 26



211P
160
210L_t
210t
210V_t
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110_R2
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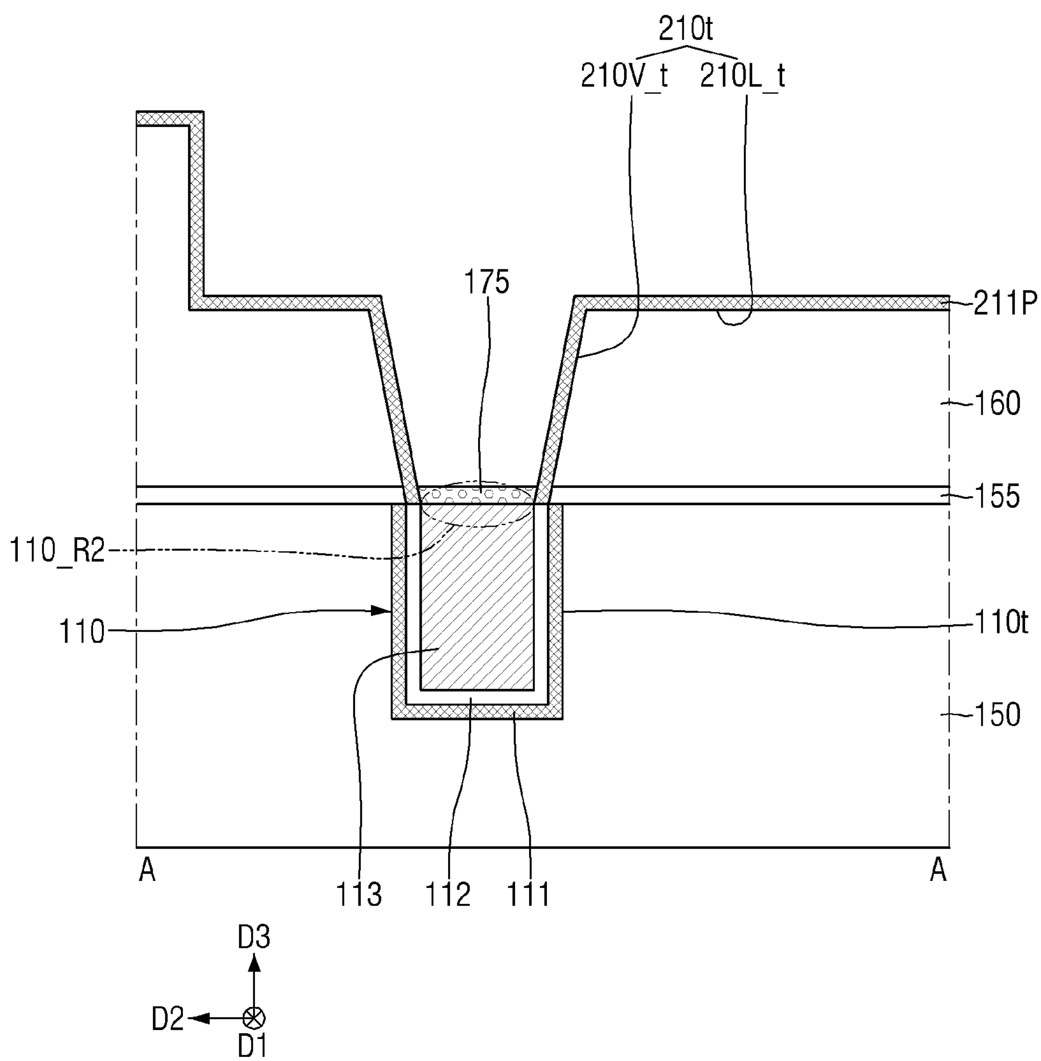
112
111
110t
150
B
B
D3
D1
D2

FIG. 27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210L_t
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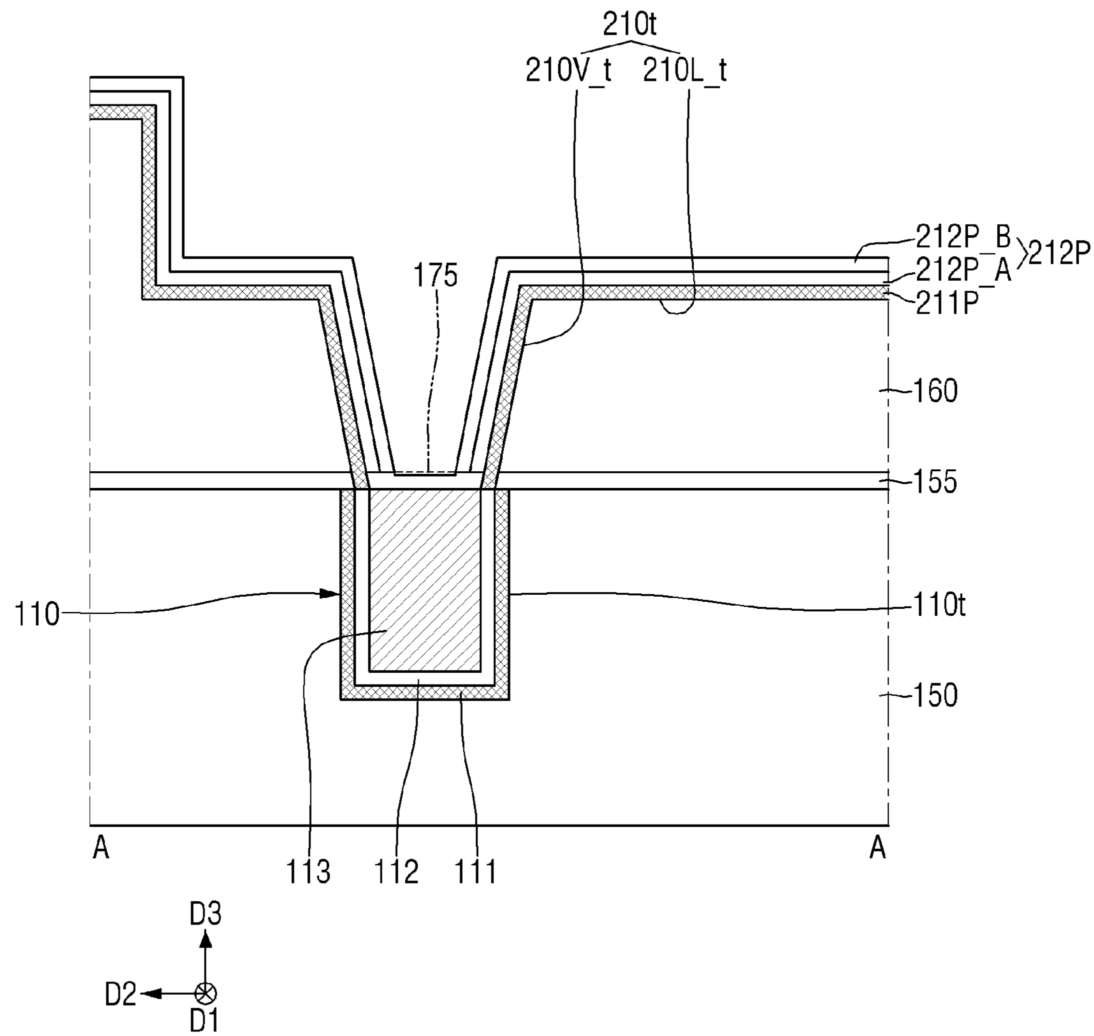
D2
D1

FIG. 28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212P_B
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212P
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A
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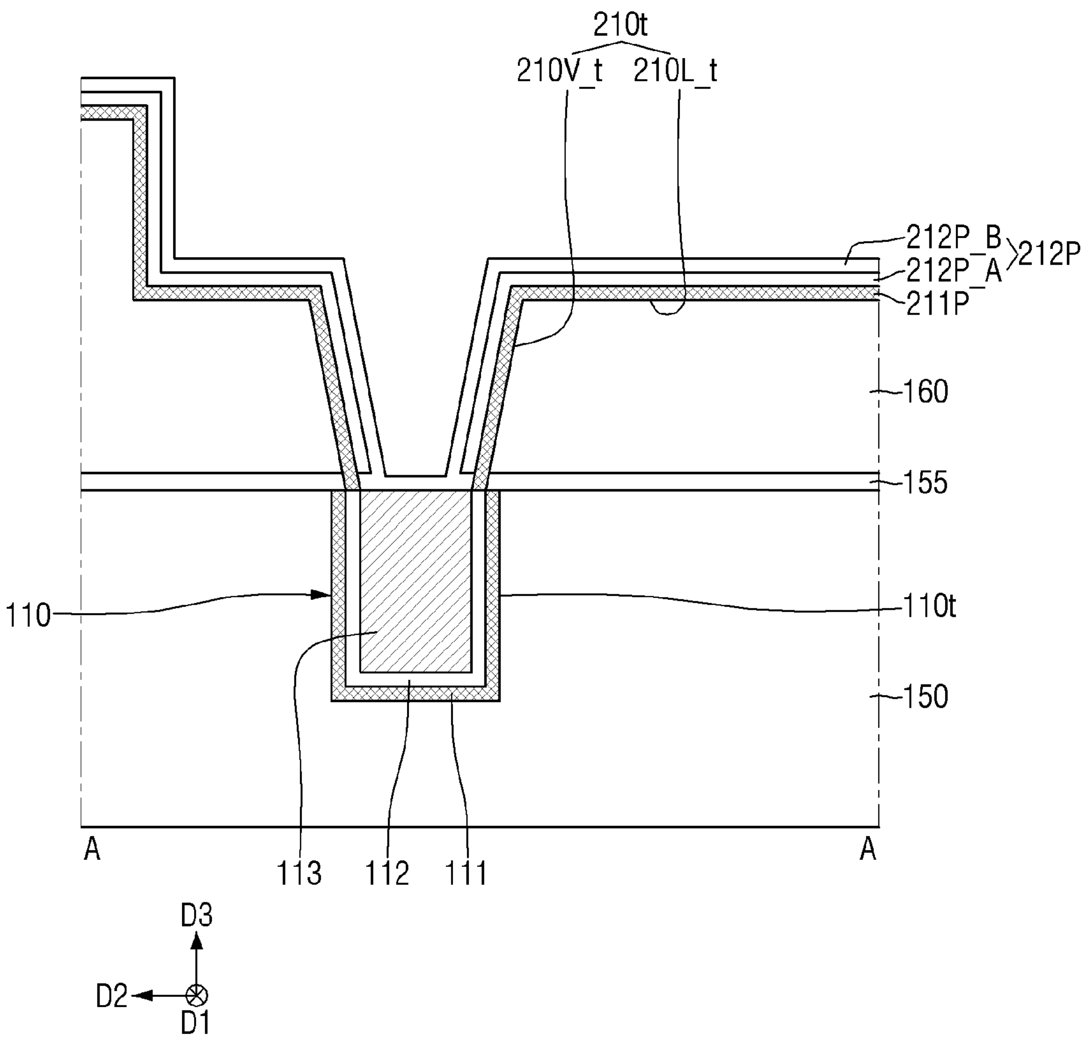
112
111
D3
D2
D1

FIG. 29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155
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110t
150
A
A
113
112
111
D3
D2
D1

SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Korean Patent Application No. 10-2021-0169761 filed on Dec. 1, 2021 in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which in its entirety are herein incorporated by reference.

FIELD

The present disclosure relates to a semiconductor device and a method for fabricating the same, and more particularly, to a semiconductor element including a wiring line formed in a back-end-of-line (BEOL) process, and a method for fabricating the same.

BACKGROUND

Due to the development of electronic technology, as down-scaling of semiconductor devices is rapidly progressing, high integration and low power consumption of semiconductor chips may be required. In order to respond to demand for high integration and low power consumption of the semiconductor chips, feature sizes of semiconductor devices are decreasing.

On the other hand, as the feature size decreases, various studies are being conducted on a stable connection method between wirings.

SUMMARY

Aspects of the present disclosure provide a semiconductor device capable of improving performance and reliability of an element.

Aspects of the present disclosure provide a method for fabricating a semiconductor device capable of improving performance and reliability of an element.

However, aspects of the present disclosure are not restricted to those set forth herein. The above and other aspects of the present disclosure will become more apparent to one of ordinary skill in the art to which the present disclosure pertains by referencing the detailed description of the present disclosure given below.

According to an aspect of the present disclosure, there is provided a semiconductor device comprising a lower wiring structure, an upper interlayer insulating layer on the lower wiring structure and including an upper wiring trench that vertically overlaps a portion of the lower wiring structure, and an upper wiring structure including an upper liner and an upper filling layer on the upper liner in the upper wiring trench, wherein the upper liner includes a sidewall portion extending along a sidewall of the upper wiring trench and a bottom portion extending along a bottom surface of the upper wiring trench, the sidewall portion of the upper liner includes cobalt (Co) and ruthenium (Ru), and the bottom portion of the upper liner is formed of cobalt (Co).

According to another aspect of the present disclosure, there is provided a semiconductor device comprising a lower wiring structure, an upper interlayer insulating layer on the lower wiring structure and including an upper wiring trench, the upper wiring trench including an upper wiring line trench and an upper via trench on a bottom surface of the upper wiring line trench, and a bottom surface of the upper via trench comprises the lower wiring structure, and an upper wiring structure including an upper barrier layer, an upper filling layer, and an upper liner between the upper barrier layer and the upper filling layer in the upper wiring trench, wherein the upper barrier layer extends along a sidewall and a bottom surface of the upper wiring line trench and a sidewall of the upper via trench, the upper barrier layer comprises a metal nitride, the upper liner includes a sidewall portion extending along the sidewall and the bottom surface of the upper wiring line trench and along the sidewall of the upper via trench, and a bottom portion extending along a bottom surface of the upper via trench, the upper liner is in contact with the lower wiring structure, the upper liner includes cobalt (Co) and ruthenium (Ru), and the sidewall portion of the upper liner has a component or composition different from that of the bottom portion of the upper liner.

According to still another aspect of the present disclosure, there is provided a semiconductor device comprising a lower wiring structure, an upper interlayer insulating layer on the lower wiring structure and including an upper wiring trench, the upper wiring trench exposing a portion of the lower wiring structure, and an upper wiring structure including an upper liner and an upper filling layer on the upper liner in the upper wiring trench, wherein the upper liner is in contact with the lower wiring structure, the upper liner includes a sidewall portion extending along a sidewall of the upper wiring trench and a bottom portion extending along a bottom surface of the upper wiring trench, the sidewall portion of the upper liner includes a first portion including a ruthenium-cobalt (RuCo) alloy layer, and a second portion formed of cobalt (Co), the second portion of the sidewall portion of the upper liner is in contact with the lower wiring structure, and the bottom portion of the upper liner is formed of cobalt (Co).

According to still another aspect of the present disclosure, there is provided a method for fabricating a semiconductor device, the method comprising forming a lower wiring structure; forming an upper interlayer insulating layer comprising an upper wiring trench on the lower wiring structure, the upper wiring trench exposing a first region of the lower wiring structure; forming a first selective suppression layer on the first region of the lower wiring structure exposed by the upper wiring trench; forming an upper barrier layer along a sidewall of the upper wiring trench in a state in which the first selective suppression layer is formed, wherein a bottom surface of the upper wiring trench is free of the upper barrier; exposing a second region of the lower wiring structure by removing the first selective suppression layer; forming a second selective suppression layer on the second region of the lower wiring structure that was exposed; forming a ruthenium (Ru) layer on the upper barrier layer along the sidewall of the upper wiring trench in a state in which the second selective suppression layer is formed, wherein the bottom surface of the upper wiring trench is free of the ruthenium layer; forming a cobalt (Co) layer on the ruthenium layer along the sidewall and the bottom surface of the upper wiring trench in the state in which the second selective suppression layer is formed; and forming a pre upper filling layer in the upper trench after removing the second selective suppression layer.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects and features of the present disclosure will become more apparent by describing in detail example embodiments thereof with reference to the attached drawings, in which:

FIG. 1 is an example layout diagram illustrating a semiconductor device according to some example embodiments.

FIG. 2 is an example cross-sectional view taken along line A-A of FIG. 1.

FIG. 3 is an example cross-sectional view taken along line B-B of FIG. 1.

FIG. 4 is an enlarged view of part P of FIG. 2.

FIG. 5 is an enlarged view of part Q of FIG. 3.

FIG. 6 is a view illustrating a semiconductor device according to some example embodiments.

FIG. 7 is a view illustrating a semiconductor device according to some example embodiments.

FIG. 8 is a view illustrating a semiconductor device according to some example embodiments.

FIG. 9 is a view illustrating a semiconductor device according to some exemplary embodiments.

FIG. 10 is a view illustrating a semiconductor device according to some example embodiments.

FIGS. 11 and 12 are views illustrating a semiconductor device according to some example embodiments.

FIGS. 13 and 14 are views illustrating a semiconductor device according to some example embodiments.

FIG. 15 is a view illustrating a semiconductor device according to some example embodiments.

FIG. 16 is a view illustrating a semiconductor device according to some example embodiments.

FIG. 17 is a view illustrating a semiconductor device according to some example embodiments.

FIGS. 18, 19, and 20 are views illustrating a semiconductor device according to some example embodiments.

FIGS. 21, 22, 23, 24, 25, 26, 27 28, and 29 are intermediate step views illustrating a method of fabricating a semiconductor device according to some example embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

In the drawings of a semiconductor device according to some example embodiments, for example, a fin-type transistor (FinFET) including a fin-type pattern-shaped channel region, a transistor including a nanowire or a nanosheet, a multi-bridge channel field effect transistor (MBCFET™), or a vertical FET is illustrated, but the present disclosure is not limited thereto. The semiconductor device according to some example embodiments may include a tunneling FET or a three-dimensional (3D) transistor. The semiconductor device according to some example embodiments may include a planar transistor. In addition, a technical idea of the present disclosure may be applied to 2D material based FETs and a heterostructure thereof.

In addition, the semiconductor device according to some example embodiments may also include a bipolar junction transistor, a lateral double-diffused metal oxide semiconductor (LDMOS) transistor, or the like.

FIG. 1 is an example layout diagram illustrating a semiconductor device according to some example embodiments. FIG. 2 is an example cross-sectional view taken along line A-A of FIG. 1. FIG. 3 is an example cross-sectional view taken along line B-B of FIG. 1. FIG. 4 is an enlarged view of part P of FIG. 2. FIG. 5 is an enlarged view of part Q of FIG. 3.

Referring to FIGS. 1 to 5, the semiconductor device according to some example embodiments may include a lower wiring structure 110 and an upper wiring structure 210.

The lower wiring structure 110 may be disposed in a first interlayer insulating layer 150. The lower wiring structure 110 may extend to be elongated in a first direction D1. The terms first, second, etc., may be used herein merely to distinguish one element or layer from another.

The lower wiring structure 110 may have a line shape extending in the first direction D1. For example, the first direction D1 may be a longitudinal direction of the lower wiring structure 110, and a second direction D2 may be a width direction of the lower wiring structure 110. Here, the first direction D1 intersects the second direction D2 and a third direction D3. The second direction D2 intersects the third direction D3.

The first interlayer insulating layer 150 may cover a gate electrode and a source/drain of a transistor formed in a front-end-of-line (FEOL) process. Alternatively, the first interlayer insulating layer 150 may be an interlayer insulating layer formed in the back-end-of-line (BEOL) process.

In other words, as an example, the lower wiring structure 110 may be a contact or a contact wiring formed in a middle-of-line (MOL) process. As another example, the lower wiring structure 110 may be a connection wiring formed in the back-end-of-line (BEOL) process. In the following description, the lower wiring structure 110 will be described as the connection wiring formed in the BEOL process.

The first interlayer insulating layer 150 may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride, or a low-k material. The low-k material may be, for example, silicon oxide with moderately high carbon and hydrogen, and may be a material such as SiCOH. Meanwhile, since carbon is included in the insulating material, a dielectric constant of the insulating material may be lowered. However, in order to further lower the dielectric constant of the insulating material, the insulating material may include pores such as gas-filled or air-filled cavities within the insulating material.

The low-k material may include, for example, Fluorinated TetraEthylOrthoSilicate (FTEOS), Hydrogen SilsesQuioxane (HSQ), Bis-benzoCycloButene (BCB), TetraMethylOrthoSilicate (TMOS), OctaMethyleyCloTetraSiloxane (OMCTS), HexaMethylDiSiloxane (HMDS), TriMethylSilyl Borate (TMSB), DiAcetoxyDitertiaryButoSiloxane (DADBS), TriMethylSilil Phosphate (TMSP), PolyTetraFluoroEthylene (PTFE), Tonen SilaZen (TOSZ), Fluoride Silicate Glass (FSG), polyimide nanofoams such as polypropylene oxide, Carbon Doped silicon Oxide (CDO), Organo Silicate Glass (OSG), SiLK, Amorphous Fluorinated Carbon, silica aerogels, silica xerogels, mesoporous silica, or a combination thereof, but is not limited thereto.

The lower wiring structure 110 may be disposed at a first metal level. The first interlayer insulating layer 150 may include a lower wiring trench 110*t* extending to be elongated in the first direction D1.

The lower wiring structure 110 may be disposed in the lower wiring trench 110*t*. The lower wiring trench 110*t* is filled with the lower wiring structure 110.

The lower wiring structure 110 may include a lower barrier layer 111, a lower liner 112, a lower filling layer 113, and a lower capping layer 114. The lower liner 112 may be disposed between the lower barrier layer 111 and the lower filling layer 113. The lower capping layer 114 may be disposed on the lower filling layer 113.

The lower barrier layer 111 may extend along sidewalls and a bottom surface of the lower wiring trench 110*t*. The lower liner 112 may be disposed on the lower barrier layer 111. The lower liner 112 may extend along the sidewalls and the bottom surface of the lower wiring trench 110*t* on the lower barrier layer 111.

The lower filling layer **113** is disposed on the lower liner **112**. The remainder of the lower wiring trench **110***t* may be filled with the lower filling layer **113**.

The lower capping layer **114** may extend along an upper surface **113**US of the lower filling layer. In some embodiments, the lower capping layer **114** may be disposed on an upper surface of the lower liner **112**. In some embodiments, unlike those illustrated in the drawings, the lower capping layer **114** may not cover the upper surface of the lower liner **112**.

When the lower liner **112** and the lower capping layer **114** are formed of the same material, the upper surface of the lower liner **112** may not be distinguished at a boundary between the lower liner **112** and the lower capping layer **114**.

In some embodiments, the lower capping layer **114** may not cover an upper surface of the lower barrier layer **111**. In some embodiments, unlike those illustrated in the drawings, the lower capping layer **114** may cover at least a portion of the upper surface of the lower barrier layer **111**.

The upper surface of the lower liner **112** is illustrated as being coplanar with the upper surface **113**US of the lower filling layer and the upper surface of the lower barrier layer **111**, but is not limited thereto. Here, the upper surface of the lower liner **112** may refer to the uppermost surface of a portion of the lower liner **112** extending along the sidewall of the lower wiring trench **110***t*.

The lower barrier layer **111** may include a conductive material, for example, metal nitride. The lower barrier layer **111** may include, for example, at least one of tantalum nitride (TaN), titanium nitride (TiN), tungsten nitride (WN), zirconium nitride (ZrN), vanadium nitride (VN), or niobium nitride (NbN). In the following description, the lower barrier layer **111** will be described as including tantalum nitride (TaN).

The lower liner **112** may include a conductive material, for example, a metal or a metal alloy. The lower liner **112** may include, for example, at least one of ruthenium (Ru), cobalt (Co), or a ruthenium-cobalt (RuCo) alloy.

In the semiconductor device according to some example embodiments, the lower liner **112** may be formed of cobalt (Co). The lower liner **112** may be formed of cobalt, and may be a lower cobalt layer **112**_BF. Here, the "cobalt layer" may be a layer formed purely of cobalt, and may include impurities introduced in a process of forming the cobalt layer. That is, when an element or layer of a device is "formed of" a material, the element or layer may substantially or entirely include the material. For example, the lower liner **112** may be formed of Co and may be free of Ru.

The lower filling layer **113** may include a conductive material, for example, at least one of aluminum (Al), copper (Cu), tungsten (W), cobalt (Co), ruthenium (Ru), silver (Ag), gold (Au), manganese (Mn), molybdenum (Mo), rhodium (Rh), iridium (Ir), RuAl, NiAl, NbB_2, MoB_2, TaB_2, V_2AlC, or CrAlC. In the semiconductor device according to some example embodiments, the lower filling layer **113** may include copper (Cu).

The lower capping layer **114** may include a conductive material, for example, a metal. The lower capping layer **114** may include, for example, at least one of cobalt (Co), ruthenium (Ru), or manganese (Mn). In the semiconductor device according to some example embodiments, the lower capping layer **114** may include cobalt. The lower capping layer **114** may be formed of cobalt (Co).

In some embodiments, unlike those illustrated in the drawings, the lower wiring structure **110** may have a single-layer structure. Although not illustrated, a via pattern connecting the conductive patterns disposed under the lower wiring structure **110** may be further included.

The lower wiring structure **110** may be formed using, for example, a damascene method. In FIG. **2**, a width of the lower wiring structure **110** in the second direction D**2** is illustrated as being constant, but is not limited thereto. In some embodiments, unlike that illustrated in FIG. **2**, as a distance from the upper surface of the first interlayer insulating layer **150** increases, the width of the lower wiring structure **110** in the second direction D**2** may decrease.

A first etch stop layer **155** may be disposed on the lower wiring structure **110** and the first interlayer insulating layer **150**. A second interlayer insulating layer **160** may be disposed on the first etch stop layer **155**. The first etch stop layer **155** may be disposed between the first interlayer insulating layer **150** and the second interlayer insulating layer **160**.

The second interlayer insulating layer **160** may include an upper wiring trench **210***t*. The upper wiring trench **210***t* may pass through the first etch stop layer **155**. The upper wiring trench **210***t* exposes a portion of the lower wiring structure **110**. The upper wiring trench **210***t* vertically overlaps a portion of the lower wiring structure **110** (e.g., in direction D**3**).

The upper wiring trench **210***t* may include an upper via trench **210**V_t and an upper wiring line trench **210**L_t. The upper wiring line trench **210**L_t may extend to be elongated in the second direction D**2**. The upper wiring line trench **210**L_t may extend into the upper surface of the second interlayer insulating layer **160**. The upper via trench **210**V_t may be formed on a bottom surface of the upper wiring line trench **210**L_t.

For example, a bottom surface of the upper wiring trench **210***t* may be a bottom surface of the upper via trench **210**V_t. The bottom surface of the upper wiring trench **210***t* may be defined by the lower wiring structure **110**.

Sidewalls of the upper wiring trench **210***t* may include sidewalls and a bottom surface of the upper wiring line trench **210**L_t and sidewalls of the upper via trench **210**V_t. The sidewalls and the bottom surface of the upper wiring line trench **210**L_t may be defined by the second interlayer insulating layer **160**. The sidewalls of the upper via trench **210**V_t may be defined by the second interlayer insulating layer **160** and the first etch stop layer **155**.

In the semiconductor device according to some example embodiments, the upper wiring trench **210***t* may pass through the lower capping layer **114**. The upper wiring trench **210***t* exposes a portion of the upper surface **113**US of the lower filling layer. In this case, a portion of the sidewall of the upper via trench **210**V_t may be defined by the lower capping layer **114**. The bottom surface of the upper wiring trench **210***t* may be defined by the upper surface **113**US of the lower filling layer.

The second interlayer insulating layer **160** may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride, or a low-k material.

The first etch stop layer **155** may include a material having an etch selectivity with respect to the second interlayer insulating layer **160**. The first etch stop layer **155** may include, for example, at least one of silicon nitride (SiN), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon boronitride (SiBN), silicon oxyboronitride (SiOBN), silicon oxycarbide (SiOC), aluminum oxide (AlO), aluminum nitride (AlN), aluminum oxycarbide (AlDC), or combinations thereof.

It has been illustrated that the first etch stop layer **155** is a single layer, but this is only for convenience of explanation and the present disclosure is not limited thereto. In some embodiments, unlike that illustrated in the drawing, the first etch stop layer **155** may include a plurality of insulating layers sequentially stacked on the first interlayer insulating layer **150**.

The upper wiring structure **210** may be disposed in the upper wiring trench **210***t*. The upper wiring trench **210***t* may be filled with the upper wiring structure **210**. The upper wiring structure **210** may be disposed in the second interlayer insulating layer **160**.

The upper wiring structure **210** is disposed on the lower wiring structure **110**. The upper wiring structure **210** is electrically connected to the lower wiring structure **110**. The upper wiring structure **210** is in contact with the lower wiring structure **110**. Elements or layers that are described as "in contact" or "in contact with" other elements or layers may refer to direct physical contact, with no intervening elements or layers therebetween.

The upper wiring structure **210** includes an upper wiring line **210**L and an upper via **210**V. The upper via **210**V electrically connects the upper wiring line **210**L and the lower wiring structure **110**. The upper via **210**V is in contact with the lower wiring structure **110**.

In the semiconductor device according to some example embodiments, the upper surface **113**US of the lower filling layer may include a first region **113**US_R1 in contact with the upper wiring structure **210** and a second region **113**US_R2 not in contact with the upper wiring structure **210**. The upper via **210**V is in contact with the first region **113**US_R1 of the upper surface of the lower filling layer.

For example, the lower capping layer **114** is disposed on the second region **113**US_R2 of the upper surface of the lower filling layer. The lower capping layer **114** covers the second region **113**US_R2 of the upper surface of the lower filling layer. The lower capping layer **114** is not disposed on the first region **113**US_R1 of the upper surface of the lower filling layer. The lower capping layer **114** does not cover the first region **113**US_R1 of the upper surface of the lower filling layer.

When a portion of the lower capping layer **114** is etched to expose the first region **113**US_R1 of the upper surface of the lower filling layer, the lower filling layer **113** is not removed by an etching process. The first region **113**US_R1 of the upper surface of the lower filling layer is coplanar with the second region **113**US_R2 of the upper surface of the lower filling layer.

The upper via trench **210**V_t and the upper wiring line trench **210**L_t are filled with the upper wiring structure **210**. The upper wiring line **210**L is disposed in the upper wiring line trench **210**L_t. The upper via **210**V is disposed in the upper via trench **210**V_t.

The upper wiring line **210**L is disposed at a second metal level different from the first metal level. The upper wiring line **210**L is disposed at a second metal level higher than the first metal level.

The upper wiring structure **210** includes an upper barrier layer **211**, an upper liner **212**, and an upper filling layer **213**. Although not illustrated, the upper wiring structure **210** may include an upper capping layer such as or similar to the lower capping layer **114**.

The upper barrier layer **211** extends along the sidewalls of the upper wiring trench **210***t*. The upper barrier layer **211** does not extend along the bottom surface of the upper wiring trench **210***t*. The upper barrier layer **211** does not cover the entirety of the lower wiring structure **110** exposed by the upper via trench **210**V_t.

The upper barrier layer **211** extends along the sidewalls and the bottom surface of the upper wiring line trench **210**L_t and the sidewalls of the upper via trench **210**V_t. The upper barrier layer **211** extends to the lower wiring structure **110** defining the bottom surface of the upper wiring trench **210***t*.

In the semiconductor device according to some example embodiments, the upper barrier layer **211** extends to the upper surface **113**US of the lower filling layer. The upper barrier layer **211** is in contact with the upper surface **113**US of the lower filling layer. In addition, the upper barrier layer **211** is in contact with the lower capping layer **114** defining the sidewall of the upper via trench **210**V_t.

The upper barrier layer **211** may include a conductive material, for example, metal nitride. The upper barrier layer **211** may include, for example, at least one of tantalum nitride (TaN), titanium nitride (TiN), tungsten nitride (WN), zirconium nitride (ZrN), vanadium nitride (VN), or niobium nitride (NbN). In the following description, it will be described that the upper barrier layer **211** is formed of tantalum nitride (TaN).

The upper liner **212** is disposed on the upper barrier layer **211**. The upper liner **212** is disposed between the upper barrier layer **211** and the upper filling layer **213**. For example, the upper liner **212** may be in contact with the upper barrier layer **211**.

The upper liner **212** extends along the sidewalls and the bottom surface of the upper wiring trench **210***t*. The upper liner **212** extends along the sidewalls and the bottom surface of the upper wiring line trench **210**L_t and the sidewalls and the bottom surface of the upper via trench **210**V_t.

The upper liner **212** is in contact with the lower wiring structure **110**. The upper liner **212** is in contact with the upper surface **113**US of the lower filling layer. The upper liner **212** extends along the upper surface **113**US of the lower filling layer.

The upper liner **212** is not in contact with the lower capping layer **114**. More specifically, the upper liner **212** is not in contact with the lower capping layer **114** defining the sidewall of the upper via trench **210**V_t.

The upper liner **212** includes a sidewall portion **212**S and a bottom portion **212**B.

The sidewall portion **212**S of the upper liner extends along the sidewall of the upper wiring trench **210***t*. The sidewall portion **212**S of the upper liner extends along the sidewalls and the bottom surface of the upper wiring line trench **210**L_t and the sidewalls of the upper via trench **210**V_t.

The sidewall portion **212**S of the upper liner extends to the lower wiring structure **110**. The sidewall portion **212**S of the upper liner is in contact with the upper surface **113**US of the lower filling layer defining the bottom surface of the upper via trench **210**V_t.

The bottom portion **212**B of the upper liner extends along the bottom surface of the upper wiring trench **210***t*. The bottom portion **212**B of the upper liner extends along the bottom surface of the upper via trench **210**V_t.

The bottom portion **212**B of the upper liner is in contact with the lower wiring structure **110**. In the semiconductor device according to some example embodiments, the bottom portion **212**B of the upper liner is in contact with the upper surface **113**US of the lower filling layer defining the bottom surface of the upper wiring trench **210***t*.

The upper liner **212** includes ruthenium (Ru) and cobalt (Co). For example, the upper liner **212** may include an upper ruthenium-cobalt (RuCo) alloy layer **212**_AF and an upper cobalt layer **212**_BF. Here, the "ruthenium-cobalt alloy layer" may be a layer formed purely of ruthenium and cobalt (and may include impurities introduced in a process of forming the alloy layer).

In the semiconductor device according to some example embodiments, the upper ruthenium-cobalt alloy layer **212**_AF extends along a portion of the sidewall of the upper wiring trench **210***t*.

The upper ruthenium-cobalt alloy layer **212**_AF extends along the sidewalls and the bottom surface of the upper wiring line trench **210**L_t. The upper ruthenium-cobalt alloy layer **212**_AF extends along a portion of the sidewall of the upper via trench **210**V_t. The upper ruthenium-cobalt alloy layer **212**_AF does not extend to the upper surface **113**US of the lower filling layer. The upper ruthenium-cobalt alloy layer **212**_AF is not in contact with the lower filling layer **113**.

For example, the upper ruthenium-cobalt alloy layer **212**_AF is in contact with the upper barrier layer **211**. The upper ruthenium-cobalt alloy layer **212**_AF is in contact with the upper filling layer **213**.

The upper cobalt layer **212**_BF extends along the bottom surface of the upper wiring trench **210***t*. In the semiconductor device according to some example embodiments, the upper cobalt layer **212**_BF is in contact with the upper barrier layer **211**. The upper cobalt layer **212**_BF is in contact with the lower filling layer **113**.

The sidewall portion **212**S of the upper liner includes ruthenium (Ru) and cobalt (Co). The bottom portion **212**B of the upper liner is formed of cobalt (Co). For example, the composition of the sidewall portion **212**S of the upper liner is different from the composition of the bottom portion **212**B of the upper liner.

The bottom portion **212**B of the upper liner is formed as a portion of the upper cobalt layer **212**_BF. The sidewall portion **212**S of the upper liner includes the remainder of the upper cobalt layer **212**_BF and the upper ruthenium-cobalt alloy layer **212**_AF.

A portion of the sidewall portion **212**S of the upper liner is formed of ruthenium (Ru) and cobalt (Co). The sidewall portion **212**S of the upper liner includes a first portion **212**S_**P1** formed of ruthenium (Ru) and cobalt (Co) and a second portion **212**S_**P2** formed of cobalt (Co). The first portion **212**S_**P1** of the sidewall portion of the upper liner is disposed on the second portion **212**S_**P2** of the sidewall portion of the upper liner. The second portion **212**S_**P2** of the sidewall portion of the upper liner is in contact with the lower wiring structure **110**.

The first portion **212**S_**P1** of the sidewall portion of the upper liner includes the upper ruthenium-cobalt alloy layer **212**_AF. For example, the first portion **212**S_**P1** of the sidewall portion of the upper liner is formed of the upper ruthenium-cobalt alloy layer **212**_AF. The second portion **212**S_**P2** of the sidewall portion of the upper liner is formed of the upper cobalt layer **212**_BF.

The upper filling layer **213** is disposed on the upper liner **212**. The upper filling layer **213** may be in contact with the upper liner **212**. The remainder of the upper wiring trench **210***t* may be filled with the upper filling layer **213**.

FIG. **6** is a view illustrating a semiconductor device according to some example embodiments. FIG. **7** is a view illustrating a semiconductor device according to some example embodiments. FIG. **8** is a view illustrating a semiconductor device according to some example embodiments. FIG. **9** is a view illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described. For reference, FIGS. **6** to **9** are enlarged views of part P of FIG. **2**.

Referring to FIG. **6**, in the semiconductor device according to some example embodiments, the upper cobalt layer **212**_BF may be disposed along a boundary between the upper ruthenium-cobalt alloy layer **212**_AF and the upper filling layer **213**.

A portion of the upper cobalt layer **212**_BF may extend along the sidewalls of the upper via trench **210**V_t and the sidewalls and the bottom surface of the upper wiring line trench **210**L_t.

The upper ruthenium-cobalt alloy layer **212**_AF may not be in contact with the upper filling layer **213**. That is, the upper ruthenium-cobalt alloy layer **212**_AF may be separated from the upper filling layer **213** by a portion of the upper cobalt layer **212**_BF extending therebetween.

Referring to FIGS. **7** and **8**, in the semiconductor device according to some example embodiments, the sidewall portion **212**S of the upper liner may further include an upper ruthenium layer **212**_CF extending along the sidewall of the upper wiring trench **210***t*.

The upper ruthenium layer **212**_CF extends along the sidewalls and the bottom surface of the upper wiring line trench **210**L_t. The upper ruthenium layer **212**_CF may extend along a portion of the sidewall of the upper via trench **210**V_t. The upper ruthenium layer **212**_CF may be in contact with the upper barrier layer **211**.

The upper ruthenium layer **212**_CF may not extend to the upper surface **113**US of the lower filling layer. The upper ruthenium layer **212**_CF may not be in contact with the lower filling layer **113**.

The upper ruthenium-cobalt alloy layer **212**_AF is disposed between the upper ruthenium layer **212**_CF and the upper filling layer **213**. The upper ruthenium layer **212**_CF is disposed between the upper ruthenium-cobalt alloy layer **212**_AF and the upper barrier layer **211**.

Here, the "ruthenium layer" may be a layer formed purely of ruthenium (and may include impurities introduced in a process of forming the ruthenium layer).

In FIG. **7**, the upper ruthenium-cobalt alloy layer **212**_AF may be in contact with the upper filling layer **213**.

In FIG. **8**, the upper ruthenium-cobalt alloy layer **212**_AF may not be in contact with the upper filling layer **213**. A portion of the upper cobalt layer **212**_BF may be disposed along a boundary between the upper ruthenium-cobalt alloy layer **212**_AF and the upper filling layer **213**.

Referring to FIG. **9**, in the semiconductor device according to some example embodiments, the entirety of the sidewall portion **212**S of the upper liner is formed of ruthenium (Ru) and cobalt (Co).

The upper ruthenium-cobalt alloy layer **212**_AF extends to the upper surface **113**US of the lower filling layer. The upper ruthenium-cobalt alloy layer **212**_AF is in contact with the lower filling layer **113**. The upper cobalt layer **212**_BF is not interposed between the upper ruthenium-cobalt alloy layer **212**_AF and the lower filling layer **113**.

The sidewall portion **212**S of the upper liner formed of ruthenium (Ru) and cobalt (Co) is in contact with the lower wiring structure **110**.

In FIGS. **7** and **8**, when the upper ruthenium-cobalt alloy layer **212**_AF extends to the upper surface **113**US of the lower filling layer, the upper ruthenium layer **212**_CF may extend to the upper surface **113**US of the lower filling layer. That is, the upper ruthenium layer **212**_CF may be in contact with the lower filling layer **113**.

FIG. **10** is a view illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described. For reference, FIG. **10** is an enlarged view of part Q of FIG. **3**.

Referring to FIG. **10**, in the semiconductor device according to some example embodiments, the upper barrier layer **211** may include a dimple region **211**_DP.

The dimple region **211**_DP may be a portion protruding between the lower surface of the first etch stop layer **155** and the upper surface of the lower filling layer **113** facing each other.

In the fabricating process, a portion of the lower capping layer **114** may be removed using wet etching to expose the lower filling layer **113**. During wet etching, the lower capping layer **114** may be under-cut. An undercut region of the lower capping layer **114** may be filled with the upper barrier layer **211**.

FIGS. **11** and **12** are views illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described.

Referring to FIGS. **4**, **11**, and **12**, in the semiconductor device according to some example embodiments, the upper liner **212** is in contact with the lower capping layer **114**. The bottom portion **212**B of the upper liner is in contact with the lower capping layer **114**.

The upper wiring trench **210***t* does not pass through the lower capping layer **114**. The upper wiring trench **210***t* does not expose the upper surface **113**US of the lower filling layer.

The sidewall of the upper via trench **210**V_t is not defined by the lower capping layer **114**. The bottom surface of the upper wiring trench **210***t* may be defined by the lower capping layer **114**.

The lower capping layer **114** may include a lower surface facing the lower filling layer **113** and an upper surface facing the first etch stop layer **155**. The upper barrier layer **211** is in contact with the upper surface of the lower capping layer **114**.

FIGS. **13** and **14** are views illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described.

Referring to FIGS. **13** and **14**, in the semiconductor device according to some example embodiments, the lower liner **112** may include cobalt (Co) and ruthenium (Ru).

The lower liner **112** may include the lower ruthenium-cobalt alloy layer **112**_AF. As an example, the entirety of the lower liner **112** may be formed of the lower ruthenium-cobalt alloy layer **112**_AF. As another example, the lower liner **112** may include a ruthenium layer as illustrated in FIGS. **7** and **8**. As still another example, the lower liner **112** may include a cobalt layer disposed along a boundary between the lower ruthenium-cobalt alloy layer **112**_AF and the lower filling layer **113**.

FIG. **15** is a view illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described.

Referring to FIG. **15**, in the semiconductor device according to some example embodiments, as the lower wiring structure **110** moves away from the upper surface of the first interlayer insulating layer **150**, the width of the lower wiring structure **110** in the second direction D**2** may increase.

The lower wiring structure **110** may be formed using, for example, a subtractive process. In other words, after a conductive layer serving as a base material of the lower wiring structure **110** is formed, a mask pattern is formed on the conductive layer. By using the mask pattern as a mask, the conductive layer is etched. Through this, the lower wiring structure **110** may be formed.

The lower wiring structure **110** is illustrated as including the lower barrier layer **111** and the lower filling layer **113**, but is not limited thereto. As an example, unlike that illustrated, a hard mask pattern may be disposed along the upper surface of the lower filling layer **113**. As another example, unlike that illustrated, a passivation layer may be disposed along the sidewall of the lower filling layer **113**. As still another example, the lower barrier layer **111** may be omitted. As still another example, the lower barrier layer **111** may include, for example, at least one of a metal nitride, a metal, a metal carbide, or a two-dimensional (2D) material. The two-dimensional material may be a metallic material and/or a semiconductor material. The 2D material may include a two-dimensional allotrope or a two-dimensional compound, and may include, for example, at least one of graphene, molybdenum disulfide (MoS_2), molybdenum diselenide ($MoSe_2$), tungsten diselenide (WSe_2), or tungsten disulfide (WS_2), but is not limited thereto. That is, since the above-described 2D material is only listed as an example, the 2D material that may be included in the semiconductor device of the present disclosure is not limited by the above-described material.

FIG. **16** is a view illustrating a semiconductor device according to some example embodiments. For convenience of explanation, points different from those described with reference to FIGS. **1** to **5** will be mainly described.

For reference, FIG. **16** exemplarily illustrates a semiconductor device cut along a first gate electrode GE.

It is illustrated in FIG. **16** that the fin-shaped pattern AF extends in the first direction D**1** and the first gate electrode GE extends in the second direction D**2**, but the present disclosure is not limited thereto.

Referring to FIG. **16**, the semiconductor device according to some example embodiments may include a transistor TR disposed between a substrate **10** and the lower wiring structure **110**.

The substrate **10** may be a silicon substrate or a silicon-on-insulator (SOI). Alternatively, the substrate **10** may include silicon germanium, silicon germanium on insulator (SGOI), indium antimonide, lead tellurium compound, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide, but is not limited thereto.

The transistor TR may include a fin-shaped pattern AF, a first gate electrode GE on the fin-shaped pattern AF, and a first gate insulating layer GI between the fin-shaped pattern AF and the first gate electrode GE.

Although not illustrated, the transistor TR may include source/drain patterns disposed on both sides of the first gate electrode GE.

The fin-shaped pattern AF may protrude from the substrate **10**. The fin-shaped pattern AF may extend to be elongated in the first direction D**1**. The fin-shaped pattern AF may be a portion of the substrate **10**, and may include an epitaxial layer grown from the substrate **10**. The fin-shaped pattern AF may include silicon or germanium, which is an elemental semiconductor material. In addition, the fin-shaped pattern AF may include a compound semiconductor, for example, a group IV-IV compound semiconductor or a group III-V compound semiconductor.

The group IV-IV compound semiconductor may be, for example, a binary compound or a ternary compound including two or more of carbon (C), silicon (Si), germanium (Ge), and tin (Sn), or a compound obtained by doping carbon (C), silicon (Si), germanium (Ge), and tin (Sn) with a group IV element. The group III-V compound semiconductor may be, for example, one of a binary compound, a ternary compound, or a quaternary compound formed by combining at least one of aluminum (Al), gallium (Ga), or indium (In), which are group III elements, with one of phosphorus (P), arsenic (As), and antimony (Sb), which are group V elements.

A field insulating layer **15** may be formed on the substrate **10**. The field insulating layer **15** may be formed on a portion of the sidewall of the fin-shaped pattern AF. The fin-shaped pattern AF may protrude above an upper surface of the field insulating layer **15**. The field insulating layer **15** may include, for example, an oxide film, a nitride film, an oxynitride film, or a combination thereof.

The first gate electrode GE may be disposed on the fin-shaped pattern AF. The first gate electrode GE may extend in the second direction D**2**. The first gate electrode GE may cross or intersect with the fin-shaped pattern AF.

The first gate electrode GE may include, for example, at least one of a metal, a conductive metal nitride, a conductive metal carbonitride, a conductive metal carbide, a metal silicide, a doped semiconductor material, a conductive metal oxynitride, or a conductive metal oxide.

The first gate insulating layer GI may be disposed between the first gate electrode GE and the fin-shaped pattern AF and between the first gate electrode GE and the field insulating layer **15**. The first gate insulating layer GI may include, for example, silicon oxide, silicon oxynitride, silicon nitride, or a high-k material having a dielectric constant greater than that of silicon oxide. The high-k material may include, for example, at least one of boron nitride, metal oxide, or metal silicon oxide.

The semiconductor device according to some example embodiments may include a negative capacitance (NC) FET using a negative capacitor. For example, the first gate insulating layer GI may include a ferroelectric material layer having ferroelectric characteristics and a paraelectric material layer having paraelectric characteristics.

The ferroelectric material layer may have a negative capacitance, and the paraelectric material layer may have a positive capacitance. For example, when two or more capacitors are connected to each other in series and the capacitance of each capacitor has a positive value, a total capacitance decreases as compared with a capacitance of each individual capacitor. On the other hand, when at least one of capacitances of two or more capacitors connected to each other in series has a negative value, a total capacitance may have a positive value and may be greater than an absolute value of each individual capacitance.

When the ferroelectric material layer having the negative capacitance and the paraelectric material layer having the positive capacitance are connected to each other in series, a total capacitance value of the ferroelectric material layer and the paraelectric material layer connected to each other in series may increase. A transistor including the ferroelectric material layer may have a subthreshold swing (SS) less than 60 mV/decade at room temperature, using the fact that the total capacitance value increases.

The ferroelectric material layer may have the ferroelectric characteristics. The ferroelectric material layer may include, for example, at least one of hafnium oxide, hafnium zirconium oxide, barium strontium titanium oxide, barium titanium oxide, or lead zirconium titanium oxide. Here, as an example, the hafnium zirconium oxide may be a material obtained by doping hafnium oxide with zirconium (Zr). As another example, the hafnium zirconium oxide may be a compound of hafnium (Hf), zirconium (Zr), and oxygen (O).

The ferroelectric material layer may further include a doped dopant. For example, the dopant may include at least one of aluminum (Al), titanium (Ti), niobium (Nb), lanthanum (La), yttrium (Y), magnesium (Mg), silicon (Si), calcium (Ca), cerium (Ce), dysprosium (Dy), erbium (Er), gadolinium (Gd), germanium (Ge), scandium (Sc), strontium (Sr), or tin (Sn). A type of dopant included in the ferroelectric material layer may change depending on a type of ferroelectric material included in the ferroelectric material layer.

When the ferroelectric material layer includes the hafnium oxide, the dopant included in the ferroelectric material layer may include, for example, at least one of gadolinium (Gd), silicon (Si), zirconium (Zr), aluminum (Al), or yttrium (Y).

When the dopant is aluminum (Al), the ferroelectric material layer may include 3 to 8 atomic % (at %) of aluminum. Here, a ratio of the dopant may be a ratio of aluminum to the sum of hafnium and aluminum.

When the dopant is silicon (Si), the ferroelectric material layer may include 2 to 10 at % of silicon. When the dopant is yttrium (Y), the ferroelectric material layer may include 2 to 10 at % of yttrium. When the dopant is gadolinium (Gd), the ferroelectric material layer may contain 1 to 7 at % of gadolinium. When the dopant is zirconium (Zr), the ferroelectric material layer may include 50 to 80 at % of zirconium.

The paraelectric material layer may have the paraelectric characteristics. The paraelectric material layer may include, for example, at least one of silicon oxide or metal oxide having a high dielectric constant. The metal oxide included in the paraelectric material layer may include, for example, at least one of hafnium oxide, zirconium oxide, or aluminum oxide, but is not limited thereto.

The ferroelectric material layer and the paraelectric material layer may include the same material. The ferroelectric material layer may have the ferroelectric characteristics, but the paraelectric material layer may not have the ferroelectric characteristics. For example, when the ferroelectric material layer and the paraelectric material layer include hafnium oxide, a crystal structure of the hafnium oxide included in the ferroelectric material layer is different from a crystal structure of the hafnium oxide included in the paraelectric material layer.

The ferroelectric material layer may have a thickness having the ferroelectric characteristics. The thickness of the ferroelectric material layer may be, for example, 0.5 to 10 nm, but is not limited thereto. Since a critical thickness representing the ferroelectric characteristics may change for each ferroelectric material, the thickness of the ferroelectric material layer may change depending on the ferroelectric material.

As an example, the first gate insulating layer GI may include one ferroelectric material layer. As another example, the first gate insulating layer GI may include a plurality of ferroelectric material layers spaced apart from each other. The first gate insulating layer GI may have a stacked layer structure in which a plurality of ferroelectric material layers and a plurality of paraelectric material layers are alternately stacked.

A gate capping pattern GE CAP may be disposed on the first gate electrode GE. The conductive lower wirings **110** and **120** may be disposed on the first gate electrode GE. Although it is illustrated that the conductive lower wirings **110** and **120** are not connected to the first gate electrode GE, the present disclosure is not limited thereto. One of the conductive lower wirings **110** and **120** may be connected to the first gate electrode GE.

FIG. **17** is a view illustrating a semiconductor device according to some example embodiments. For convenience of explanation, portions different from those described with reference to FIG. **16** will be mainly described.

Referring to FIG. **17**, in the semiconductor device according to some example embodiments, the transistor TR may include a nanosheet NS, a first gate electrode GE surrounding the nanosheet NS, and a first gate insulating layer GI between the nanosheet NS and the first gate electrode GE.

The nanosheet NS may be disposed on a lower fin-shaped pattern BAF. The nanosheet NS may be spaced apart from the lower fin-shaped pattern BAF in the third direction D**3**. The transistor TR is illustrated as including three nanosheets NS spaced apart from each other in the third direction D**3**, but is not limited thereto. The number of nanosheets NS disposed in the third direction D**3** on the lower fin-shaped pattern BAF may be greater than three or less than three.

Each of the lower fin-shaped pattern BAF and the nanosheet NS may include, for example, silicon or germanium, which is an elemental semiconductor material. Each of the lower fin-shaped pattern BAF and the nanosheet NS may include a compound semiconductor, for example, a group Iv-Iv compound semiconductor or a group III-v compound semiconductor. The lower fin-shaped pattern BAF and the nanosheet NS may include the same material or different materials.

FIGS. **18** to **20** are views illustrating a semiconductor device according to some example embodiments. For reference, FIG. **18** is a view illustrating a semiconductor device according to some example embodiments. FIG. **19** is a cross-sectional view taken along lines C-C and D-D of FIG. **18**. FIG. **20** is a cross-sectional view taken along line E-E of FIG. **18**.

Referring to FIGS. **18** to **20**, a logic cell LC may be provided on the substrate **10**. The logic cell LC may refer to a logic device (e.g., an inverter, a flip-flop, etc.) that performs a specific function. The logic cell LC may include vertical FETs constituting a logic device and wirings connecting the vertical FETs to each other.

The logic cell LC on the substrate **10** may include a first active region RX**1** and a second active region RX**2**. For example, the first active region RX**1** may be a PMOSFET region, and the second active region RX**2** may be an NMOSFET region. The first and second active regions RX**1** and RX**2** may be defined by a trench T_CH formed in an upper portion of the substrate **10**. The first and second active regions RX**1** and RX**2** may be spaced apart from each other in the first direction D**1**.

A first lower epitaxial pattern SPO**1** may be provided on the first active region RX**1**, and a second lower epitaxial pattern SPO**2** may be provided on the second active region RX**2**. In plan view, the first lower epitaxial pattern SPO**1** may overlap the first active region RX**1**, and the second lower epitaxial pattern SPO**2** may overlap the second active region RX**2**. The first and second lower epitaxial patterns SPO**1** and SPO**2** may be epitaxial patterns formed by a selective epitaxial growth process. The first lower epitaxial pattern SPO**1** may be provided in a first recess region RS**1** of the substrate **10**, and the second lower epitaxial pattern SPO**2** may be provided in a second recess region RS**2** of the substrate **10**.

First active patterns AP**1** may be provided on the first active region RX**1**, and second active patterns AP**2** may be provided on the second active region RX**2**. Each of the first and second active patterns AP**1** and AP**2** may have a fin shape that vertically protrudes (e.g. in direction D**3**). In plan view, each of the first and second active patterns AP**1** and AP**2** may have a bar shape extending in the first direction D**1**. The first active patterns AP**1** may be arranged along the second direction D**2**, and the second active patterns AP**2** may be arranged along the second direction D**2**.

Each of the first active patterns AP**1** may include a first channel pattern CHP**1** vertically protruding from the first lower epitaxial pattern SPO**1** and a first upper epitaxial pattern DOP**1** on the first channel pattern CHP**1**. Each of the second active patterns AP**2** may include a second channel pattern CHP**2** vertically protruding from the second lower epitaxial pattern SPO**2** and a second upper epitaxial pattern DOP**2** on the second channel pattern CHP**2**.

A element separation layer ST may be provided on the substrate **10** to fill the trench T_CH. The element separation layer ST may cover upper surfaces of the first and second lower epitaxial patterns SPO**1** and SPO**2**. The first and second active patterns AP**1** and AP**2** may vertically protrude above the element separation layer ST.

A plurality of second gate electrodes **420** extending parallel to each other in the first direction D**1** may be provided on the element separation layer ST. The second gate electrodes **420** may be arranged along the second direction D**2**. The second gate electrode **420** may surround a first channel pattern CHP**1** of the first active pattern AP**1** and may surround a second channel pattern CHP**2** of the second active pattern AP**2**. For example, the first channel pattern CHP**1** of the first active pattern AP**1** may have first to fourth sidewalls SW**1** to SW**4**. The first and second sidewalls SW**1** and SW**2** may face or oppose each other in the second direction D**2**, and the third and fourth sidewalls SW**3** and SW**4** may face or oppose each other in the first direction D**1**. The second gate electrode **420** may be provided on the first to fourth sidewalls SW**1** to SW**4**. In other words, the second gate electrode **420** may surround the first to fourth sidewalls SW**1** to SW**4**.

A second gate insulating layer **430** may be interposed between the second gate electrode **420** and each of the first and second channel patterns CHP**1** and CHP**2**. The second gate insulating layer **430** may cover a bottom surface of the second gate electrode **420** and an inner sidewall of the second gate electrode **420**. For example, the second gate insulating layer **430** may directly cover the first to fourth sidewalls SW**1** to SW**4** of the first active pattern AP**1**.

The first and second upper epitaxial patterns DOP**1** and DOP**2** may vertically protrude above the second gate electrode **420**. An upper surface of the second gate electrode **420** may be lower than a bottom surface of each of the first and second upper epitaxial patterns DOP**1** and DOP**2**. In other words, each of the first and second active patterns AP**1** and AP**2** may have a structure that vertically protrudes from the substrate **10** and penetrates the second gate electrode **420**.

The semiconductor device according to some example embodiments may include vertical transistors in which carriers move in the third direction D**3**. For example, when a voltage is applied to the second gate electrode **420** to "turn on" the transistor, the carriers may move from the lower epitaxial patterns SOP**1** and SOP**2** to the upper epitaxial patterns DOP**1** and DOP**2** through the channel patterns CHP**1** and CHP**2**. In the semiconductor device according to some example embodiments, the second gate electrode **420** may completely surround the sidewalls SW**1** to SW**4** of the channel patterns CHP1 and CHP2. The transistor according to the present disclosure may be a three-dimensional field effect transistor (e.g., VFET) having a gate all around structure. Since the gate surrounds the channel, the semiconductor device according to some example embodiments may have excellent electrical characteristics.

A spacer **440** covering the second gate electrodes **420** and the first and second active patterns AP**1** and AP**2** may be provided on the element separation layer ST. The spacer **440** may include a silicon nitride layer or a silicon oxynitride layer. The spacer **440** may include a lower spacer **440**LS, an upper spacer **440**US, and a gate spacer **440**GS between the lower and upper spacers **440**LS and **440**US.

The lower spacer **440**LS may directly cover an upper surface of the element separation layer ST. The second gate electrodes **420** may be spaced apart from the element separation layer ST in the third direction D**3** by the lower spacer **440**LS. The gate spacer **440**GS may cover an upper surface and an outer sidewall of each of the second gate electrodes **420**. The upper spacer **440**US may cover the first and second upper epitaxial patterns DOP**1** and DOP**2**. However, the upper spacer **440**US may not cover upper surfaces of the first and second upper epitaxial patterns DOP**1** and DOP**2**, but may expose the upper surfaces of the first and second upper epitaxial patterns DOP**1** and DOP**2**.

A first portion **190**BP of the lower interlayer insulating layer may be provided on the spacer **440**. An upper surface of the first portion **190**BP of the lower interlayer insulating layer may be substantially coplanar with the upper surfaces of the first and second upper epitaxial patterns DOP**1** and DOP**2**. A second portion **190**UP of the lower interlayer insulating layer and the first and second interlayer insulating layers **150** and **160** may be sequentially stacked on the first portion **190**BP of the lower interlayer insulating layer. The first portion **190**BP of the lower interlayer insulating layer and the second portion **190**UP of the lower interlayer insulating layer may be included in a lower interlayer insulating layer **190**. The second portion **190**UP of the lower interlayer insulating layer may cover the upper surfaces of the first and second upper epitaxial patterns DOP**1** and DOP**2**.

At least one first source/drain contact **470** that penetrates through the second portion **190**UP of the lower interlayer insulating layer and is connected to the first and second upper epitaxial patterns DOP**1** and DOP**2** may be provided. At least one second source/drain contact **570** that sequentially penetrates through the lower interlayer insulating layer **190**, the lower spacer **440**LS, and the element separation layer ST and is connected to the first and second lower epitaxial patterns SPO**1** and SPO**2** may be provided. A gate contact **480** that sequentially penetrates through the second portion **190**UP of the lower interlayer insulating layer, the first portion **190**BP of the lower interlayer insulating layer, and the gate spacer **440**GS and is connected to the second gate electrode **420** may be provided.

A second etch stop layer **156** may be additionally disposed between the second portion **190**UP of the lower interlayer insulating layer and the first interlayer insulating layer **150**. The first etch stop layer **155** may be disposed between the first interlayer insulating layer **150** and the second interlayer insulating layer **160**.

The lower wiring structure **110** may be provided in the first interlayer insulating layer **150**. The lower wiring structure **110** may include a lower via **110**V and a lower wiring line **110**L. Descriptions of the lower via **110**V and the lower wiring line **110**L may be similar to those of the upper via **210**V and the upper wiring line **210**L. However, the layer structure of the lower wiring structure **110** may be different from or the same as the layer structure of the upper wiring structure **210**.

The lower wiring structure **110** may be connected to the first source/drain contact **470**, the second source/drain contact **570**, and the gate contact **480**. The upper wiring structure **210** may be provided in the second interlayer insulating layer **160**.

In some embodiments, unlike that illustrated, for example, an additional wiring structure similar to the upper wiring structure **210** may be further disposed between the first source/drain contact **470** and the lower wiring structure **110**.

A detailed description of the upper wiring structure **210** may be substantially the same as that described above with reference to FIGS. **1** to **15**.

FIGS. **21** to **29** are intermediate step views illustrating a method of fabricating a semiconductor device according to some example embodiments.

For reference, FIGS. **21**, **23** to **25**, and **27** to **29** are cross-sectional views taken along line A-A of FIG. **1**, respectively. FIGS. **22** and **26** are cross-sectional views taken along line B-B of FIG. **1**.

Referring to FIGS. **21** and **22**, a lower wiring structure **110** is formed in a first interlayer insulating layer **150**.

A lower wiring trench **110***t* is formed in the first interlayer insulating layer **150**. The lower wiring structure **110** is formed in the lower wiring trench **110***t*. The lower wiring structure **110** may include a lower barrier layer **111**, a lower liner **112**, a lower filling layer **113**, and a lower capping layer **114**.

Subsequently, a first etch stop layer **155** may be formed on the first interlayer insulating layer **150** and the lower wiring structure **110**.

A second interlayer insulating layer **160** may be disposed on the first etch stop layer **155**. The second interlayer insulating layer **160** may include an upper wiring trench **210***t*. The upper wiring trench **210***t* includes an upper via trench **210**V_t and an upper wiring line trench **210**L_t.

The upper wiring trench **210***t* may penetrate through the first etch stop layer **155**. The upper wiring trench **210***t* may pass through the lower capping layer **114**. In some embodiments, unlike that illustrated, the upper wiring trench **210***t* may not pass through the lower capping layer **114**.

The upper wiring trench **210***t* may expose a first region **110**_R1 of the lower wiring structure. The upper wiring trench **210***t* may expose a portion of an upper surface of the lower filling layer **113**.

Referring to FIG. **23**, a first selective suppression layer **170** is formed on the first region **110**_R1 of the lower wiring structure exposed by the upper wiring trench **210***t*.

The first selective suppression layer **170** includes an organic material. The first selective suppression layer **170** may selectively prevent a conductive material from being deposited on a surface on which the first selective suppression layer **170** is formed. For example, a conductive material A is not deposited on the surface on which the first selective suppression layer **170** is formed. On the other hand, a conductive material B may be deposited on the surface on which the first selective suppression layer **170** is formed.

The first selective suppression layer **170** may be formed on a conductive material. For example, the first selective suppression layer **170** may be formed on a metal or a metal alloy. The first selective suppression layer **170** is not formed on an insulating material.

Referring to FIG. **24**, responsive to forming the first selective suppression layer **170** or in a state in which the first selective suppression layer **170** is formed, a pre lower barrier layer **211**P is formed along a sidewall of the upper wiring trench **210***t*. The pre lower barrier layer **211**P is formed along the upper surface of the second interlayer insulating layer **160**.

The pre lower barrier layer **211**P is not formed along a bottom surface of the upper wiring trench **210***t* in which the first selective suppression layer **170** is formed.

However, the pre lower barrier layer **211**P may be formed on the entirety of the sidewall of the upper wiring trench **210***t*. The first selective suppression layer **170** covers a portion of the sidewall of the upper wiring trench **210***t*, but the pre lower barrier layer **211**P may also be formed on the sidewall of the upper wiring trench **210***t* covered by the first selective suppression layer **170**. The pre lower barrier layer **211**P is in contact with the lower wiring structure **110**.

The pre lower barrier layer **211**P is formed using, for example, atomic layer deposition (ALD). The pre lower barrier layer **211**P may include metal nitride, for example, tantalum nitride (TaN).

Referring to FIGS. **25** and **26**, the second region **110**_**R2** of the lower wiring structure may be exposed by removing the first selective suppression layer **170**.

Since the pre lower barrier layer **211**P covers a portion of the first region **110**_**R1** of the lower wiring structure, the second region **110**_**R2** of the lower wiring structure may be smaller than the first region **110**_**R1** of the lower wiring structure.

The first selective suppression layer **170** may be removed through, for example, plasma processing, but is not limited thereto.

Referring to FIG. **27**, the second selective suppression layer **175** is formed on the exposed second region **110**_**R2** of the lower wiring structure.

A second selective suppression layer **175** includes an organic material. The second selective suppression layer **175** may selectively prevent a conductive material from being deposited on a surface on which the second selective suppression layer **175** is formed.

The second selective suppression layer **175** may be formed on a metal or a metal alloy. The second selective suppression layer **175** is not formed on an insulating material. In addition, the second selective suppression layer **175** is not formed on metal nitride having a dense structure.

Referring to FIG. **28**, responsive to forming the second selective suppression layer **175** or in a state in which the second selective suppression layer **175** is formed, a first pre upper liner **212**P_A is formed. The first pre upper liner **212**P_A is formed on the pre lower barrier layer **211**P.

The first pre upper liner **212**P_A is formed along a sidewall of the upper wiring trench **210***t*. The first pre upper liner **212**P_A is formed along the upper surface of the second interlayer insulating layer **160**.

The first pre upper liner **212**P_A is not formed along a bottom surface of the upper wiring trench **210***t* in which the second selective suppression layer **175** is formed. In addition, the first pre upper liner **212**P_A may not be formed on the sidewall of the upper wiring trench **210***t* covered by the second selective suppression layer **175**. In some embodiments, unlike that illustrated, the first pre upper liner **212**P_A may also be formed on the sidewall of the upper wiring trench **210***t* covered by the second selective suppression layer **175**.

The first pre upper liner **212**P_A may be, for example, a ruthenium (Ru) layer.

Here, in a state in which the second selective suppression layer **175** is formed, a second pre upper liner **212**P_B is formed. The second pre upper liner **212**P_B is formed on the first pre upper liner **212**P_A.

The second pre upper liner **212**P_B is formed along the sidewall and the bottom surface of the upper wiring trench **210***t*. The second pre upper liner **212**P_B is formed along the upper surface of the second interlayer insulating layer **160**.

The second pre upper liner **212**P_B is formed along the bottom surface of the upper wiring trench **210***t* in which the second selective suppression layer **175** is formed.

The second pre upper liner **212**P_B may be a cobalt (Co) layer.

In a state in which the second selective suppression layer **175** is formed, a pre upper liner **212**P is formed along the sidewall and the bottom surface of the upper wiring trench **210***t*. The pre upper liner **212**P includes the first pre upper liner **212**P_A and the second pre upper liner **212**P_B.

Referring to FIG. **29**, the second selective suppression layer **175** is removed.

The second selective suppression layer **175** may be removed through, for example, plasma processing, but is not limited thereto.

Subsequently, referring to FIG. **2**, a pre upper filling layer filling the upper wiring trench **210***t* may be formed on the pre upper liner **212**P.

Subsequently, the pre lower barrier layer **211**P, the pre upper liner **212**P, and the pre upper filling layer disposed on the upper surface of the second interlayer insulating layer **160** may be removed.

For example, while the second pre upper liner **212**P_B is formed on the first pre upper liner **212**P_A, an alloy reaction between cobalt (Co) and ruthenium (Ru) may proceed.

As another example, while the second selective suppression layer **175** is removed, an alloy reaction between cobalt (Co) and ruthenium (Ru) may proceed.

As still another example, while the pre upper filling layer fills the upper wiring trench **210***t*, an alloy reaction between cobalt (Co) and ruthenium (Ru) may proceed. When a temperature at which the alloy reaction between cobalt (Co) and ruthenium (Ru) occurs is maintained, the alloy reaction between cobalt (Co) and ruthenium (Ru) may proceed.

In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications may be made to the embodiments without substantially departing from the principles of the present inventive concept. Therefore, the disclosed embodiments of the invention are used in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A semiconductor device comprising:

a lower wiring structure;

an upper interlayer insulating layer on the lower wiring structure and comprising an upper wiring trench that vertically overlaps a portion of the lower wiring structure; and

an upper wiring structure comprising an upper liner and an upper filling layer on the upper liner in the upper wiring trench,

wherein the upper liner comprises a sidewall portion extending along a sidewall of the upper wiring trench and a bottom portion extending along a bottom surface of the upper wiring trench,

the sidewall portion of the upper liner comprises cobalt (Co) and ruthenium (Ru), and

the bottom portion of the upper liner is formed of cobalt (Co).

2. The semiconductor device of claim 1, wherein the sidewall portion of the upper liner comprises a first portion formed of cobalt (Co) and ruthenium (Ru) and a second portion formed of cobalt (Co), and

the first portion of the sidewall portion of the upper liner is on the second portion of the sidewall portion of the upper liner.

3. The semiconductor device of claim 1, wherein an entirety of the sidewall portion of the upper liner is formed of cobalt (Co) and ruthenium (Ru), and

the sidewall portion of the upper liner is in contact with the lower wiring structure.

4. The semiconductor device of claim 1, wherein the sidewall portion of the upper liner comprises a ruthenium-cobalt (RuCo) alloy layer.

5. The semiconductor device of claim 4, wherein the sidewall portion of the upper liner further comprises a cobalt (Co) layer extending along a boundary between the ruthenium-cobalt alloy layer and the upper filling layer.

6. The semiconductor device of claim 4, wherein the sidewall portion of the upper liner further comprises a ruthenium (Ru) layer extending along the sidewall of the upper wiring trench, and

the ruthenium-cobalt alloy layer is between the ruthenium layer and the upper filling layer.

7. The semiconductor device of claim 1, wherein the bottom portion of the upper liner is in contact with the lower wiring structure and is free of ruthenium (Ru).

8. The semiconductor device of claim 1, wherein the upper wiring structure further comprises an upper barrier layer extending along the sidewall of the upper wiring trench,

the upper liner is between the upper barrier layer and the upper filling layer,

the upper barrier layer comprises a metal nitride, and

the upper barrier layer does not extend along the bottom surface of the upper wiring trench.

9. The semiconductor device of claim 8, wherein the upper barrier layer is in contact with the lower wiring structure.

10. The semiconductor device of claim 1, wherein the lower wiring structure comprises a lower filling layer and a lower capping layer on the lower filling layer, and

the bottom portion of the upper liner is in contact with the lower capping layer.

11. The semiconductor device of claim 1, wherein the lower wiring structure comprises a lower filling layer and a lower capping layer on the lower filling layer,

an upper surface of the lower filling layer comprises a first region comprising the lower capping layer and a second region free of the lower capping layer, and

the upper wiring structure is in contact with the second region of the upper surface of the lower filling layer.

12. A semiconductor device comprising:

a lower wiring structure;

an upper interlayer insulating layer on the lower wiring structure and comprising an upper wiring trench, the upper wiring trench comprising an upper wiring line trench and an upper via trench on a bottom surface of the upper wiring line trench, wherein a bottom surface of the upper via trench is defined by the lower wiring structure; and

an upper wiring structure comprising an upper barrier layer, an upper filling layer, and an upper liner between the upper barrier layer and the upper filling layer in the upper wiring trench,

wherein the upper barrier layer extends along a sidewall and the bottom surface of the upper wiring line trench and a sidewall of the upper via trench,

the upper barrier layer comprises a metal nitride,

the upper liner comprises a sidewall portion extending along the sidewall and the bottom surface of the upper wiring line trench and along the sidewall of the upper via trench, and a bottom portion extending along the bottom surface of the upper via trench,

the upper liner is in contact with the lower wiring structure,

the upper liner comprises cobalt (Co) and ruthenium (Ru), and

the sidewall portion of the upper liner has a composition different from that of the bottom portion of the upper liner.

13. The semiconductor device of claim 12, wherein the sidewall portion of the upper liner comprises cobalt (Co) and ruthenium (Ru), and

the bottom portion of the upper liner is formed of cobalt (Co).

14. The semiconductor device of claim 12, wherein the sidewall portion of the upper liner comprises a first portion comprising cobalt (Co) and ruthenium (Ru) and a second portion formed of cobalt (Co), and

the second portion of the sidewall portion of the upper liner is in contact with the lower wiring structure.

15. The semiconductor device of claim 12, wherein an entirety of the sidewall portion of the upper liner comprises cobalt (Co) and ruthenium (Ru), and

the sidewall portion of the upper liner is in contact with the lower wiring structure.

16. The semiconductor device of claim 12, wherein the upper barrier layer is in contact with the lower wiring structure, and the bottom portion of the upper liner is free of ruthenium (Ru).

17. A semiconductor device comprising:

a lower wiring structure;

an upper interlayer insulating layer on the lower wiring structure and comprising an upper wiring trench that vertically overlaps a portion of the lower wiring structure; and

an upper wiring structure comprising an upper liner and an upper filling layer on the upper liner in the upper wiring trench,

wherein the upper liner is in contact with the lower wiring structure,

the upper liner comprises a sidewall portion extending along a sidewall of the upper wiring trench and a bottom portion extending along a bottom surface of the upper wiring trench,

the sidewall portion of the upper liner comprises a first portion comprising a ruthenium-cobalt (RuCo) alloy layer, and a second portion formed of cobalt (Co),

the second portion of the sidewall portion of the upper liner is in contact with the lower wiring structure, and

the bottom portion of the upper liner is formed of cobalt (Co).

18. The semiconductor device of claim 17, wherein the upper wiring structure further comprises an upper barrier layer extending along the sidewall of the upper wiring trench,

the upper liner is between the upper barrier layer and the upper filling layer,

the upper barrier layer comprises a tantalum nitride, and

the upper barrier layer is in contact with the lower wiring structure.

19. The semiconductor device of claim **17**, further comprising a lower interlayer insulating layer comprising a lower wiring trench,

wherein the lower wiring structure comprises:

a lower liner extending along a sidewall of the lower wiring trench and formed of cobalt (Co),

a lower filling layer in the lower wiring trench on the lower liner, and

a lower capping layer on an upper surface of the lower filling layer, in contact with the lower liner, and formed of cobalt (Co).

20. The semiconductor device of claim **17**, further comprising a lower interlayer insulating layer comprising a lower wiring trench,

wherein the lower wiring structure comprises:

a lower liner extending along a sidewall of the lower wiring trench and comprising cobalt (Co) and ruthenium (Ru),

a lower filling layer in the lower wiring trench on the lower liner, and

a lower capping layer on an upper surface of the lower filling layer, in contact with the lower liner, and formed of cobalt (Co).

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